

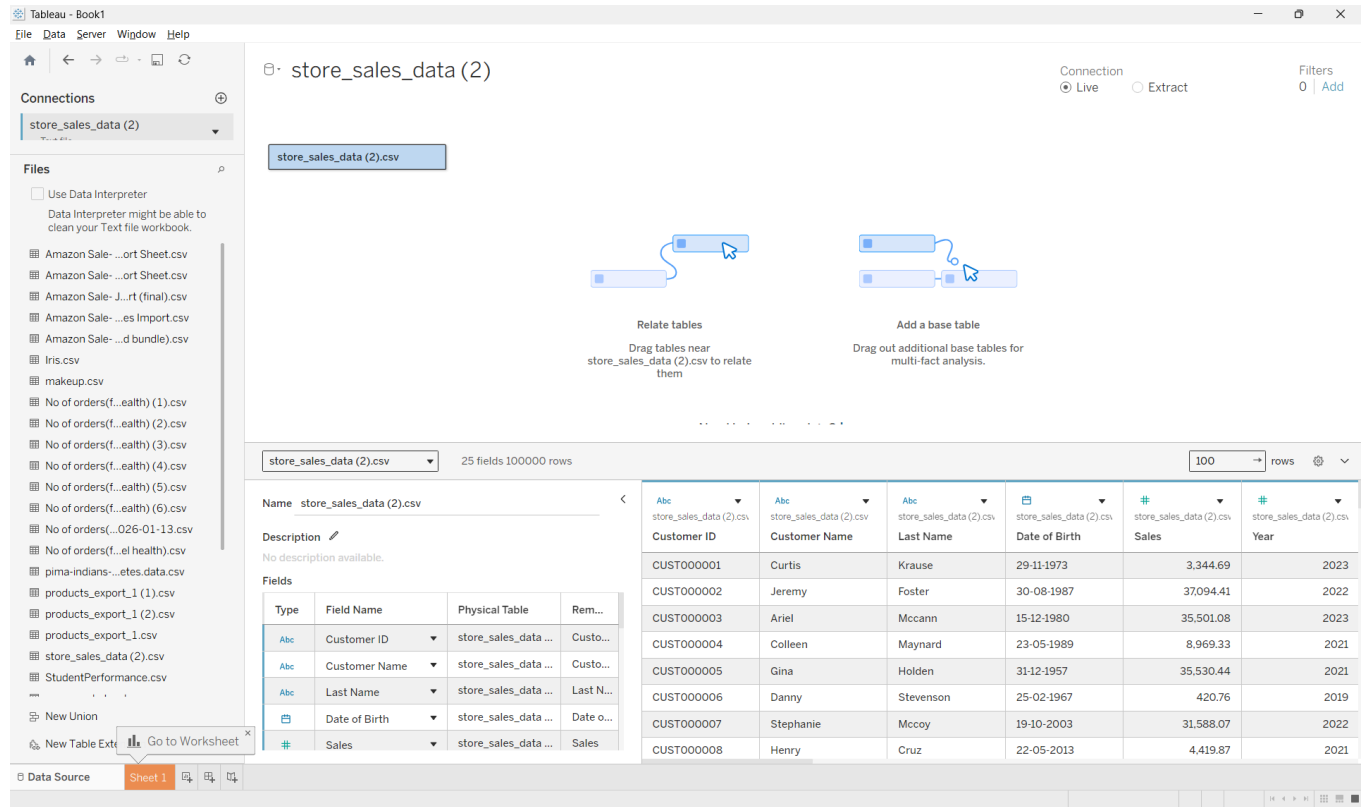
Sheth L.U.J & Sir M.V College

Subject: Data Visualization with Power BI and Tableau

Lesson 4: Performing Data Analysis using Calculations in Tableau

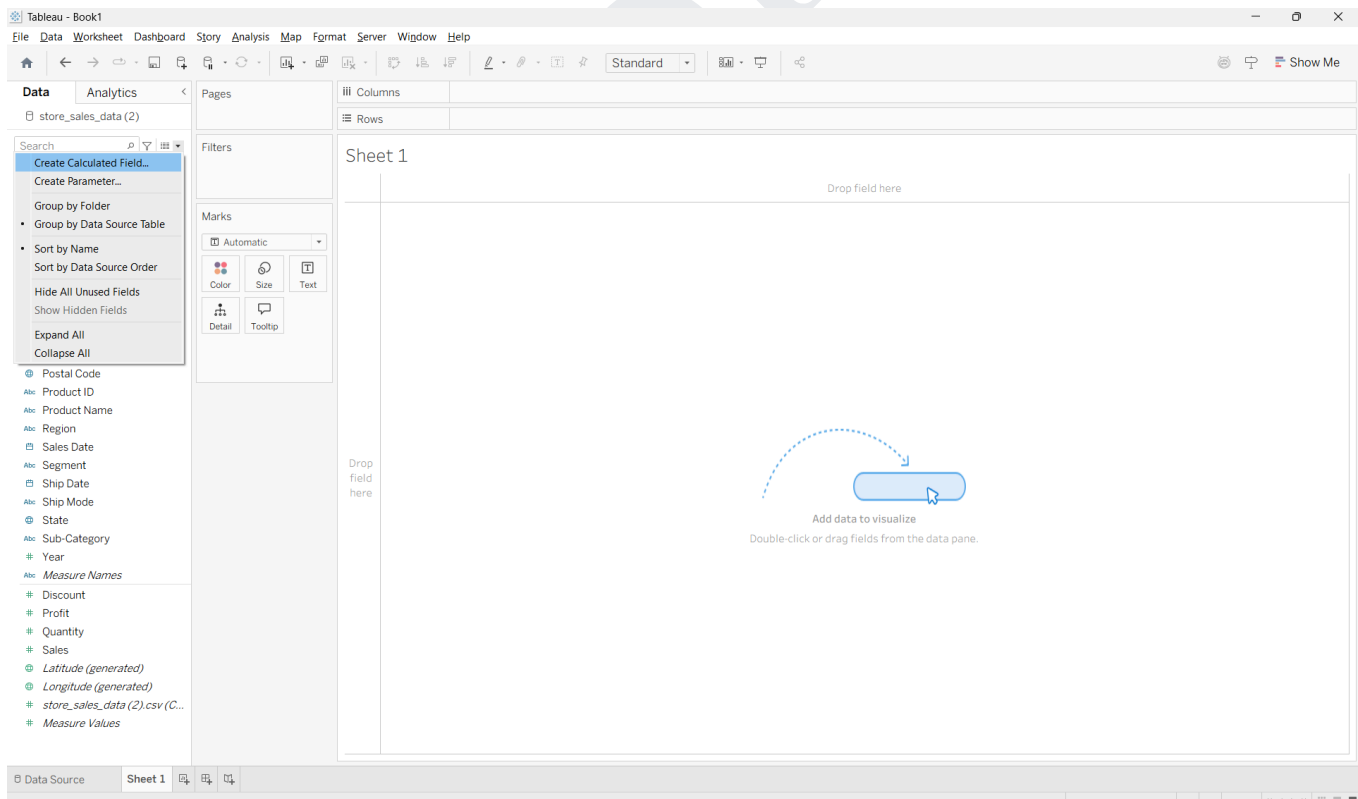
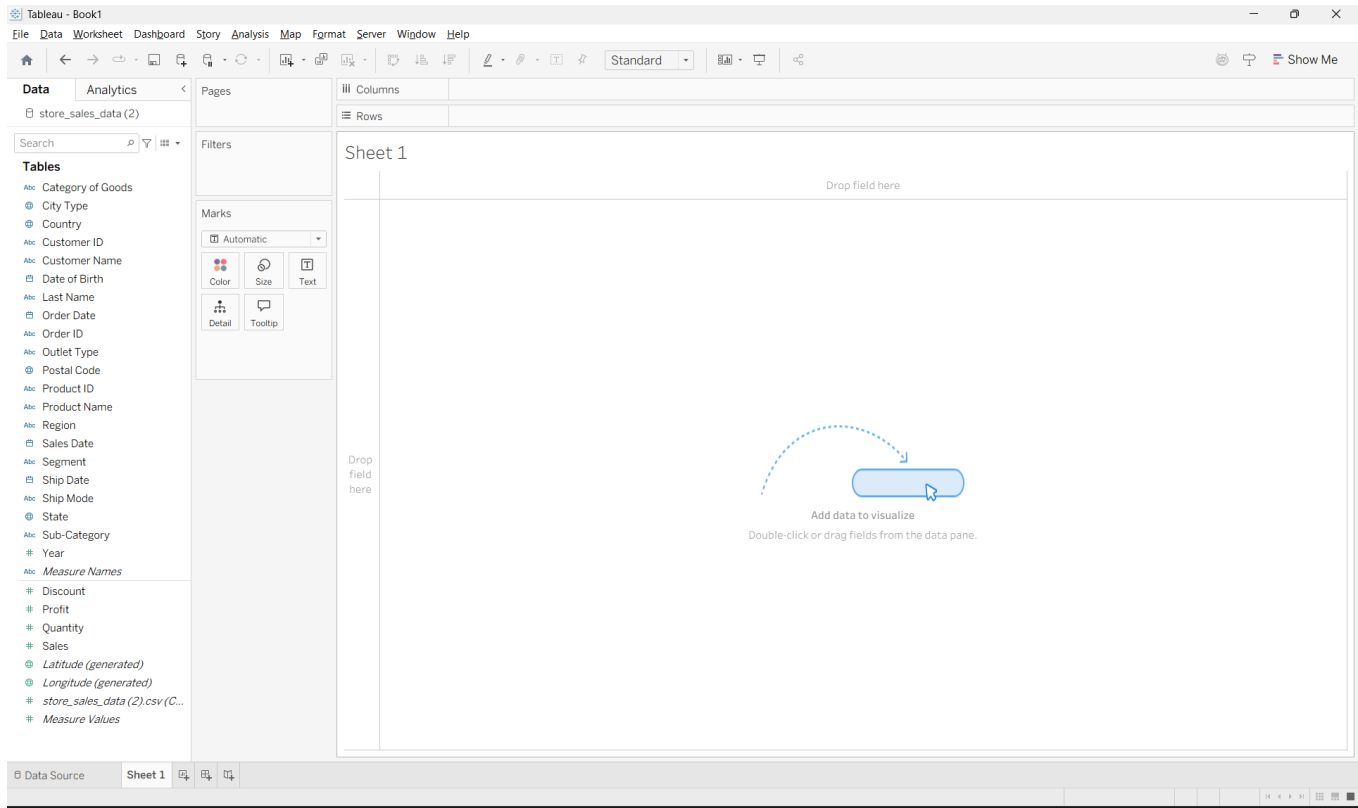
1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your `store_sales_data.csv`, and click Open.



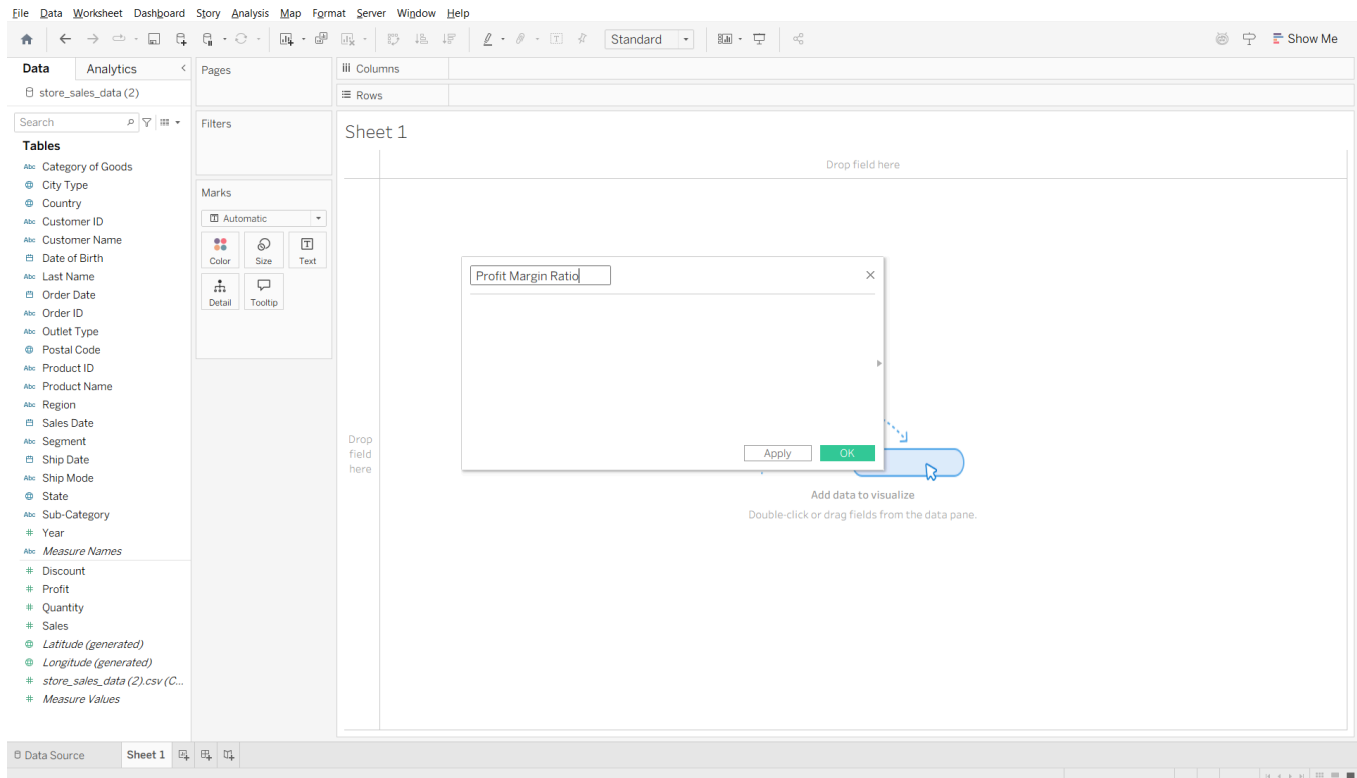
Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Click on Sheet 1 at the bottom-left corner to enter your workspace.
2. Building a Basic Custom Calculation
1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...

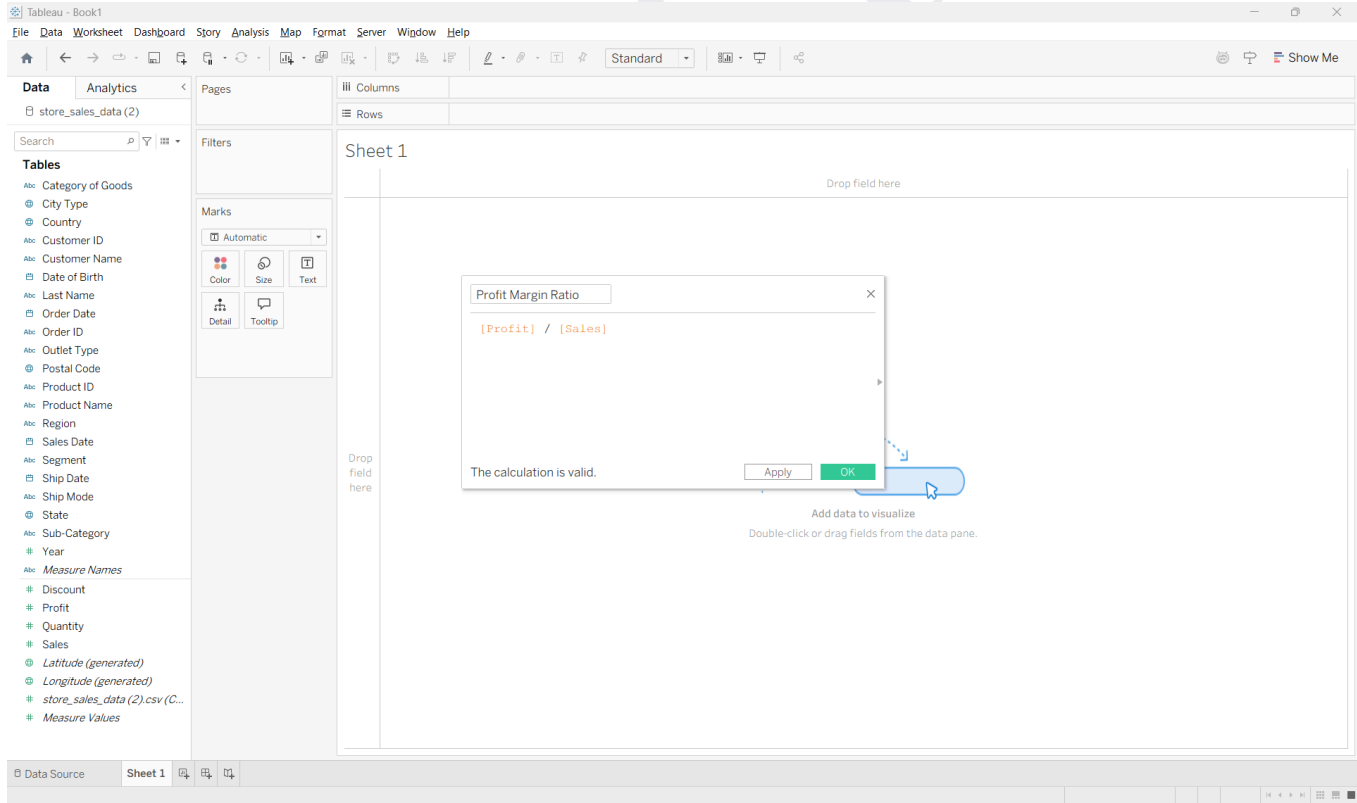


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.

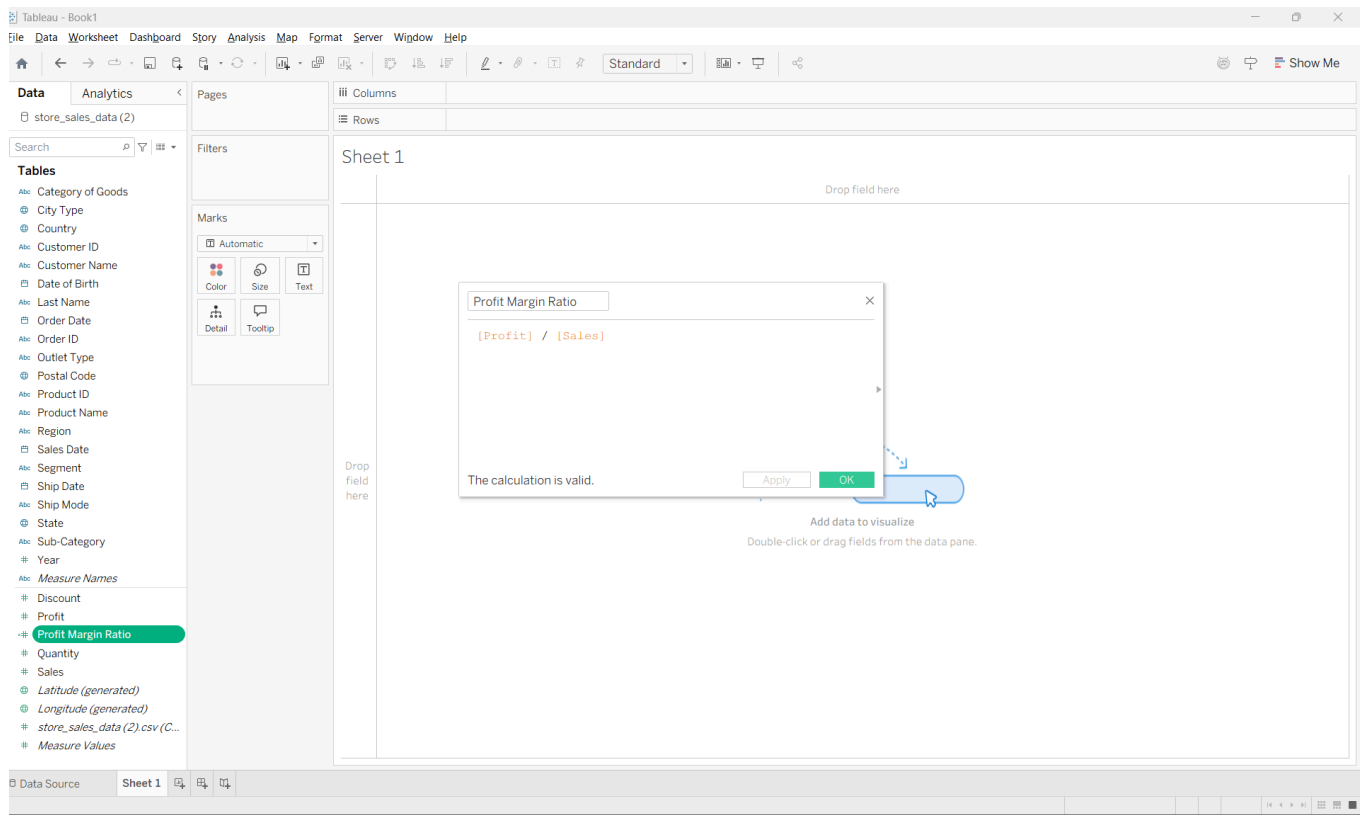


3. In the main workspace box, enter the following basic mathematical formula:
[Profit] / [Sales]

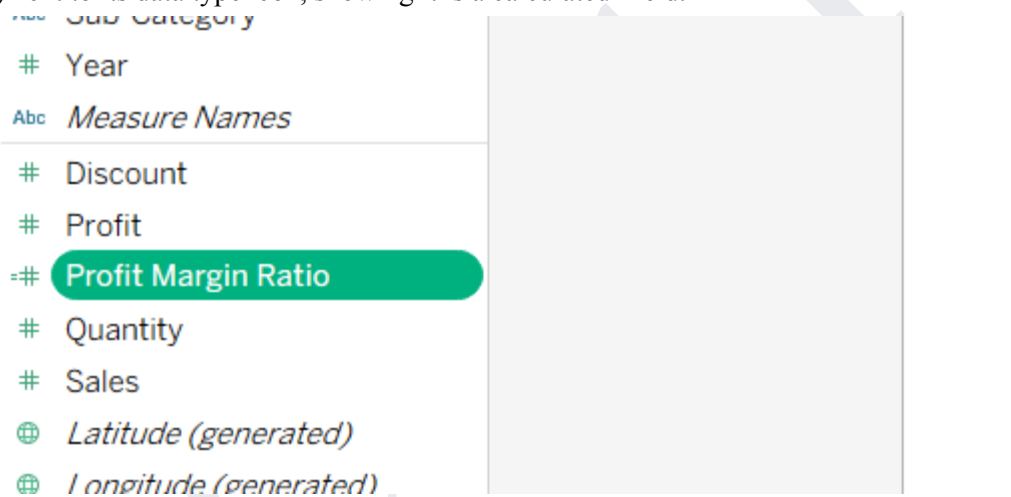


4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid." Click OK.

Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau



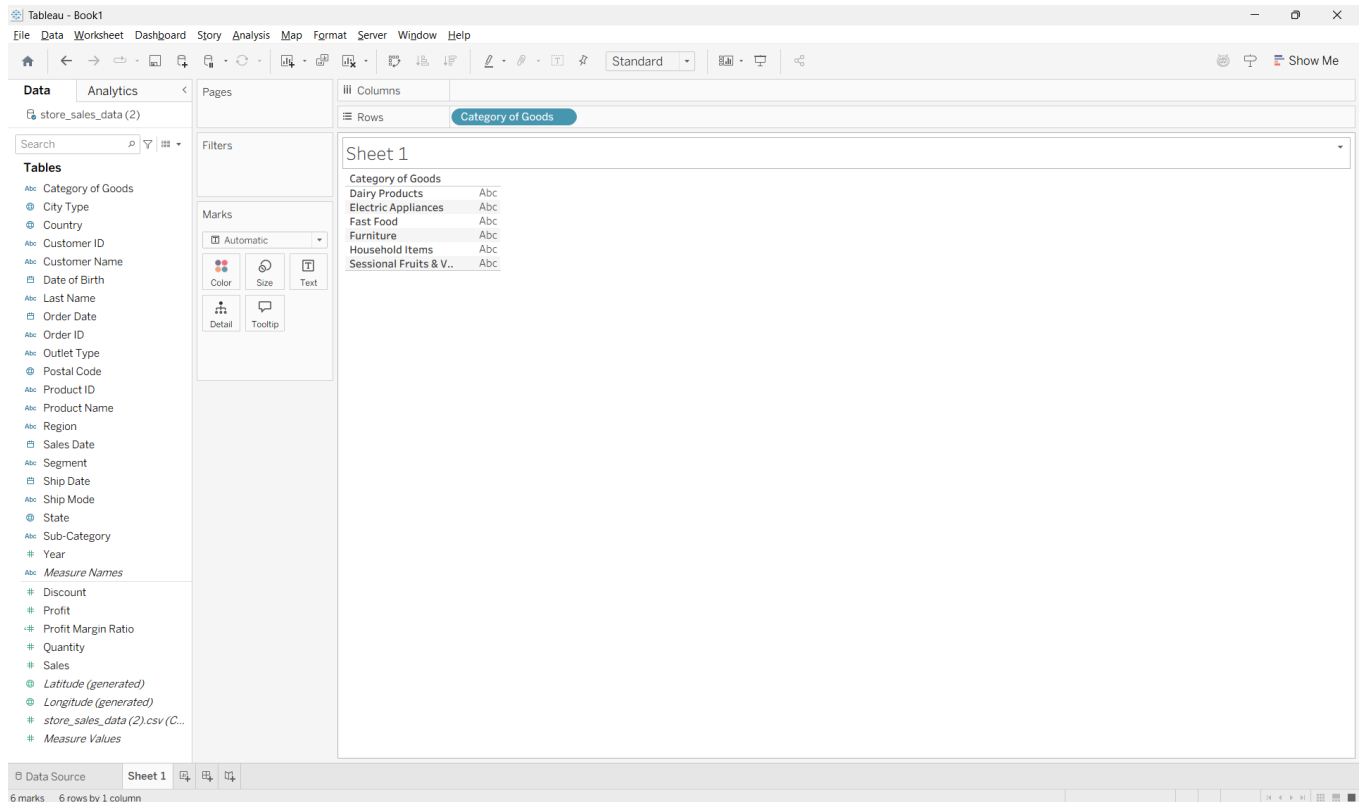
5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.



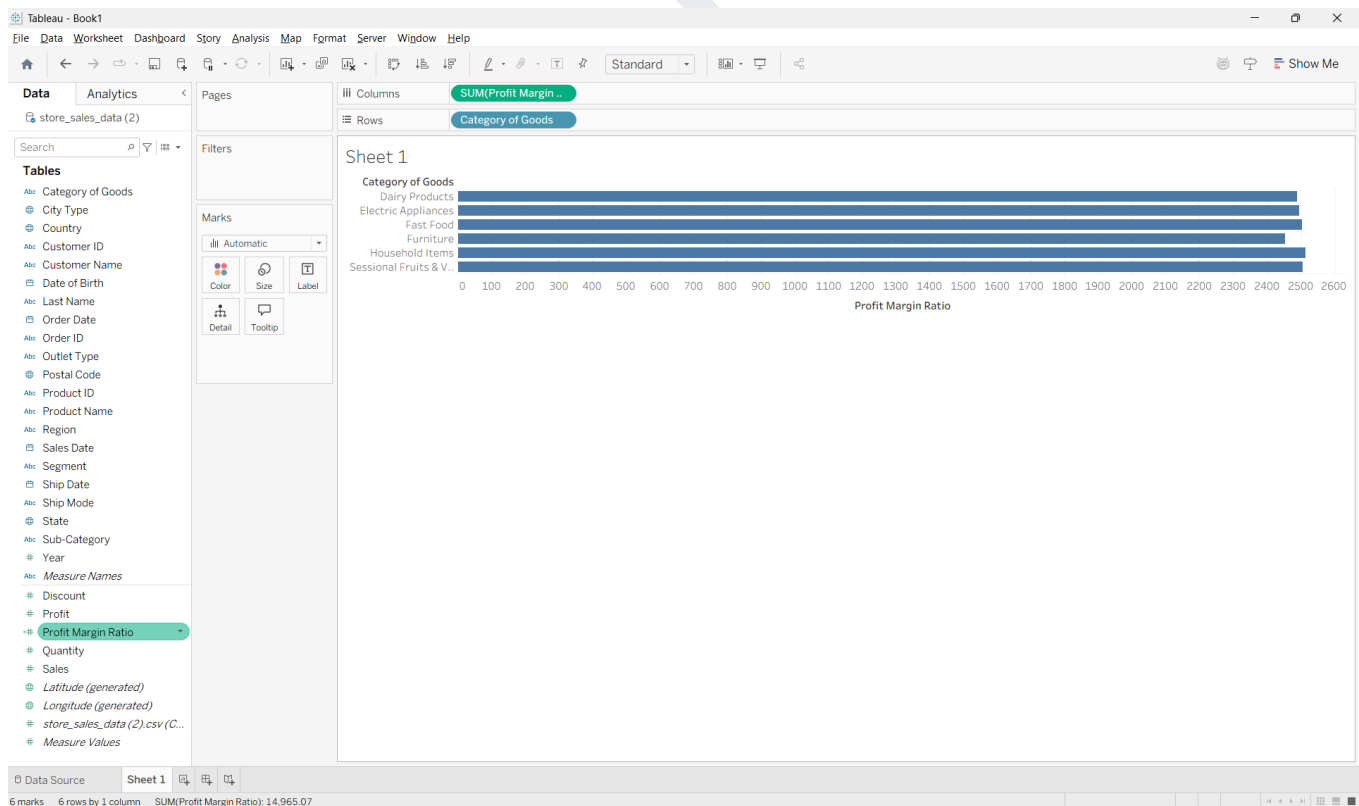
Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Putting the Field to Use

1. Drag the Category of Goods dimension from your Data Pane and drop it onto the Rows shelf.

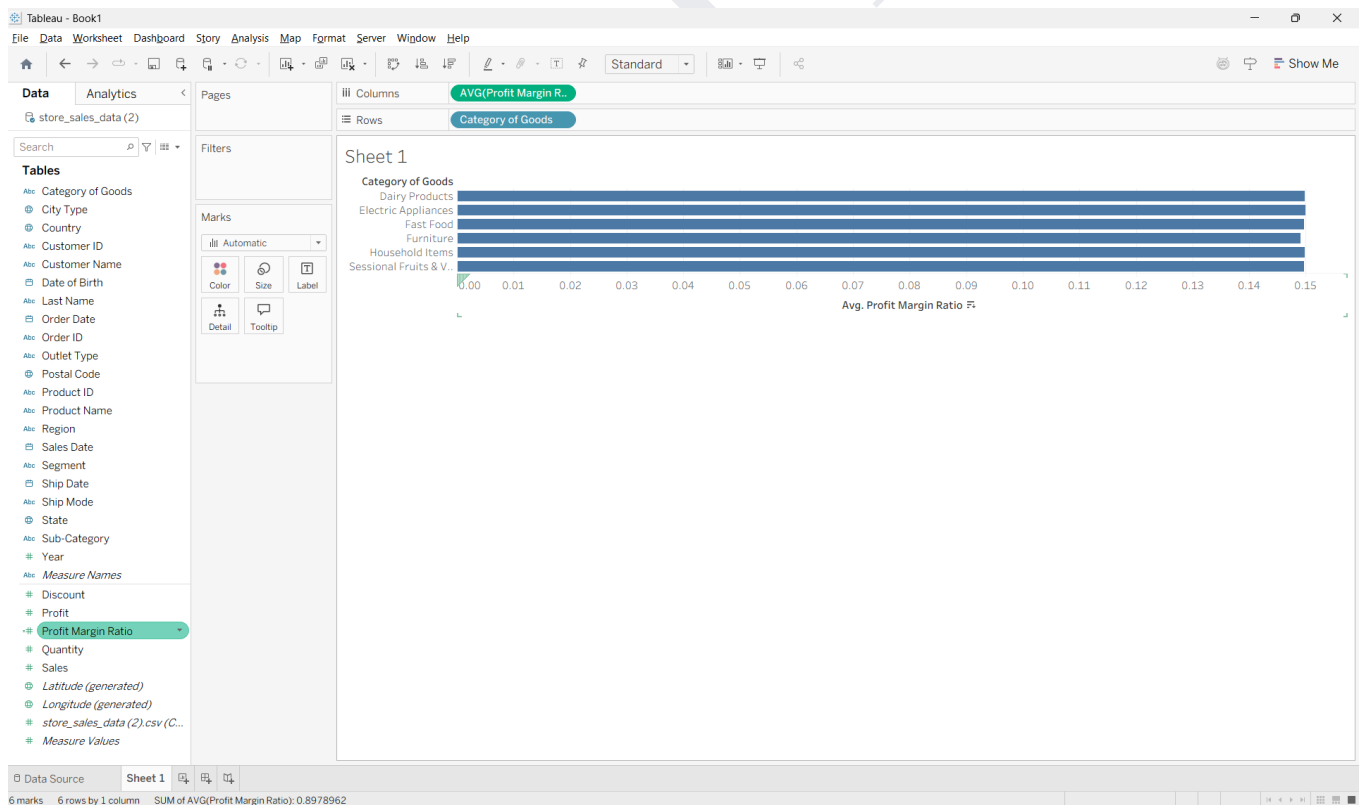
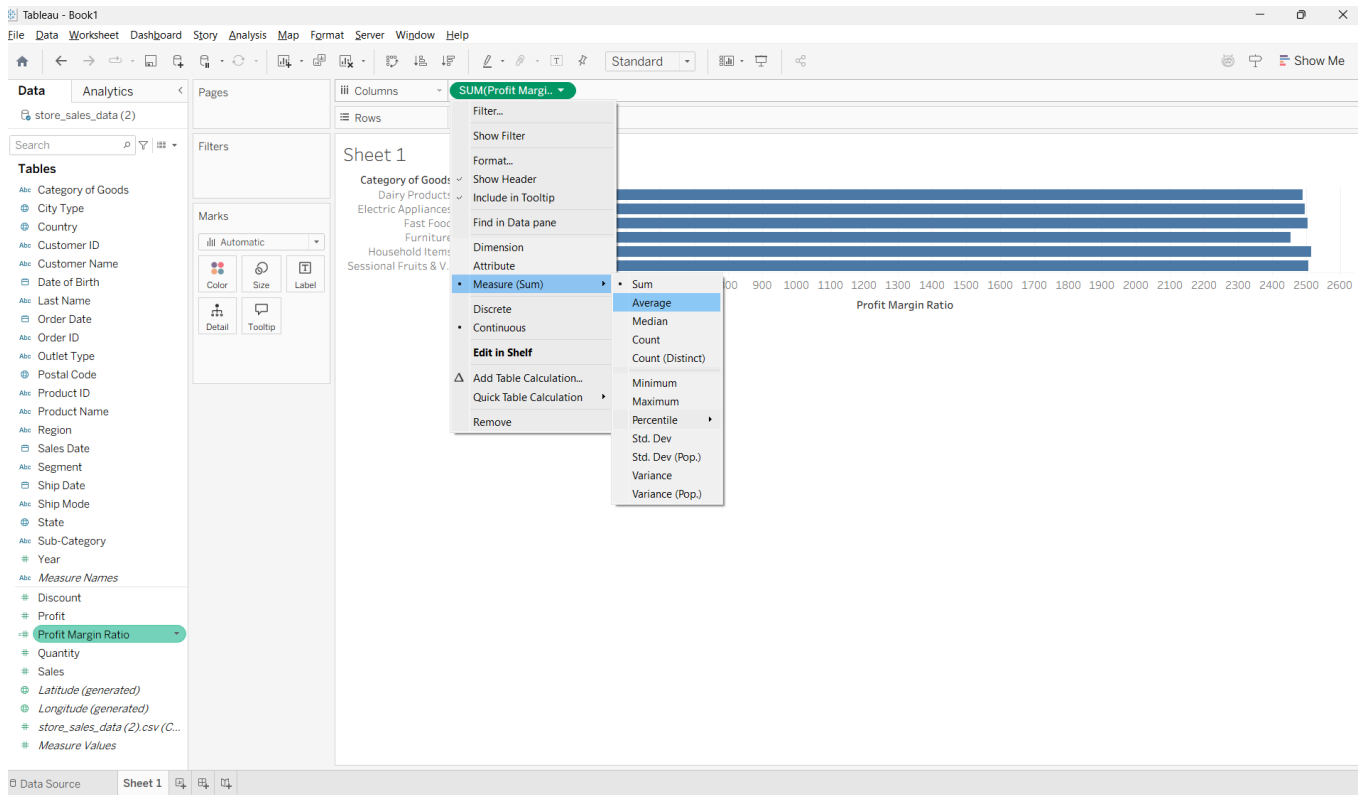


2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.



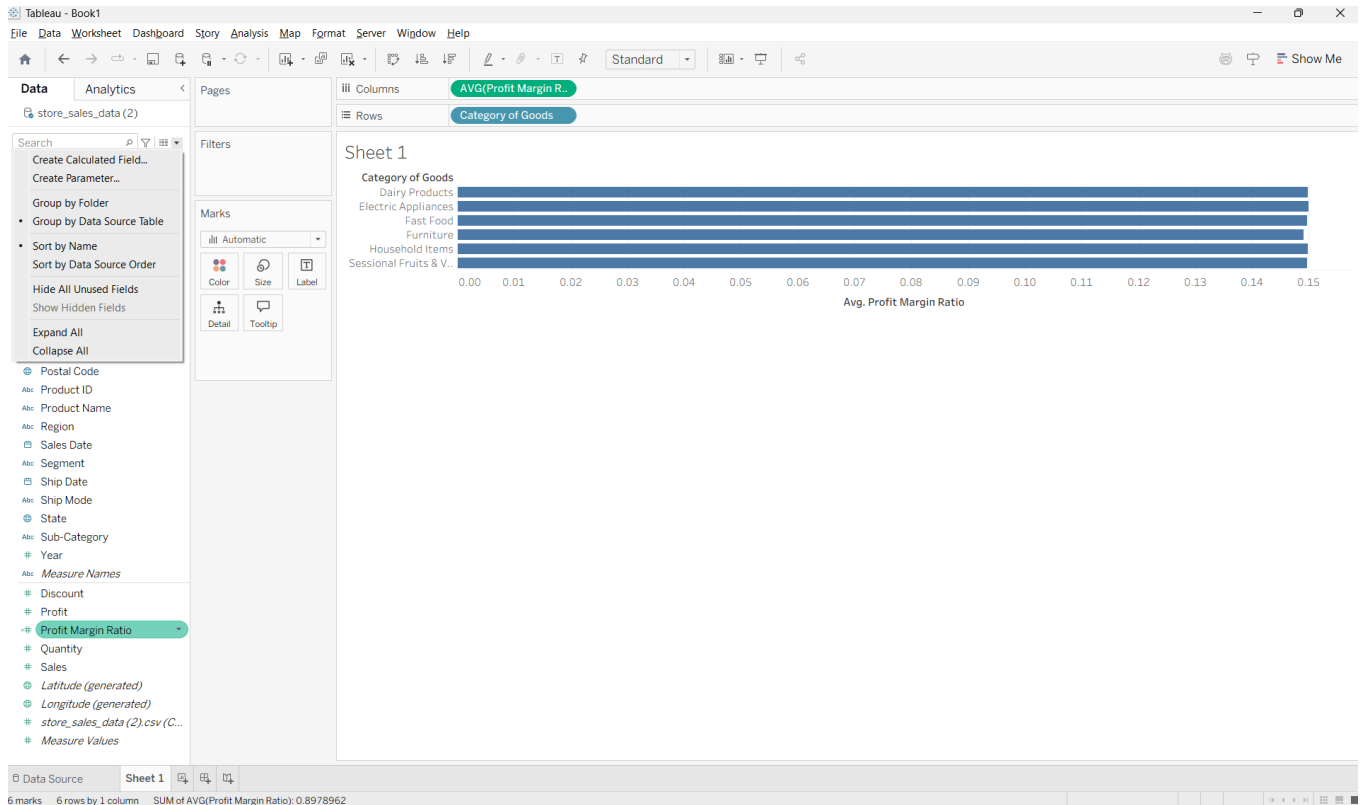
Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

Use basic formulas whenever you need to calculate financial margins, combine text strings (like merging First Name and Last Name), or perform basic arithmetic scaling on individual rows before plotting them.

B: Performing Row-Level Calculations

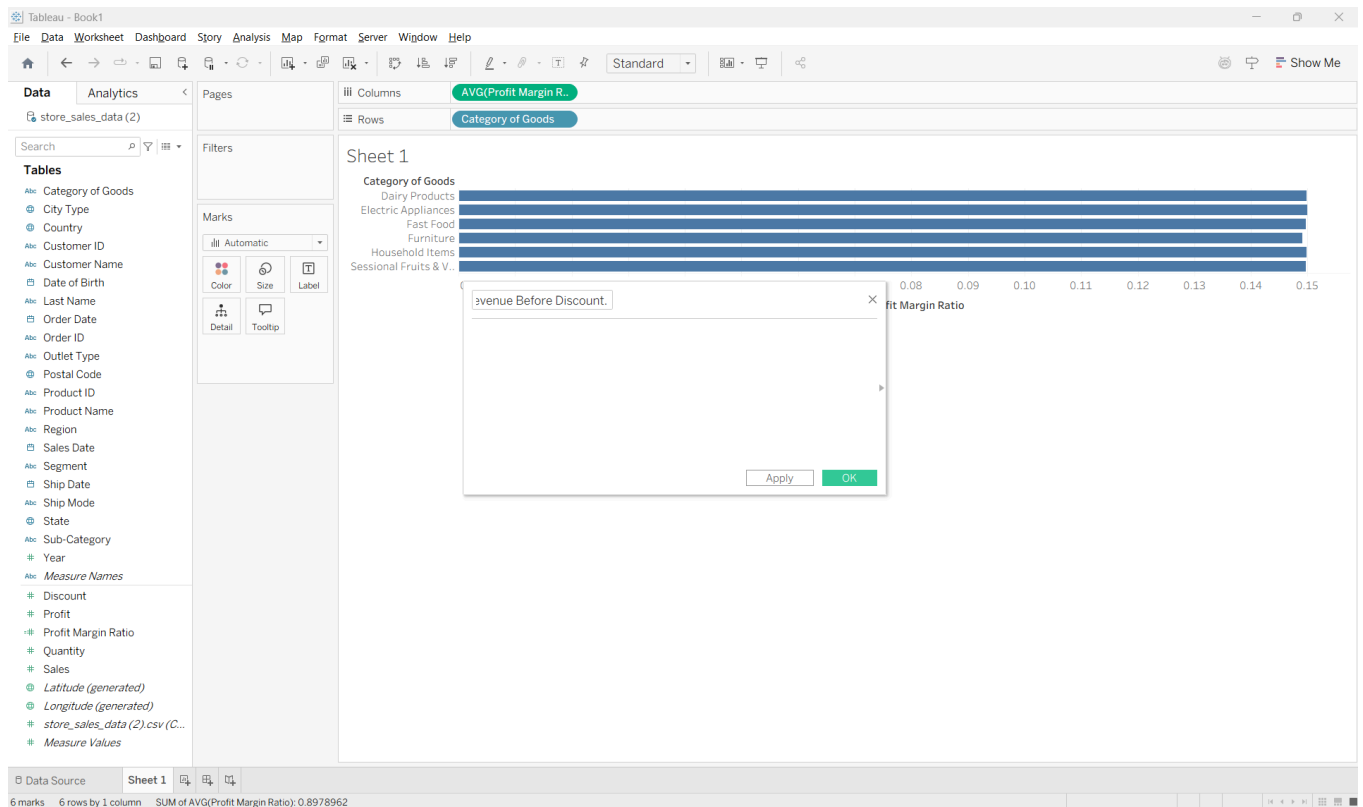
1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...

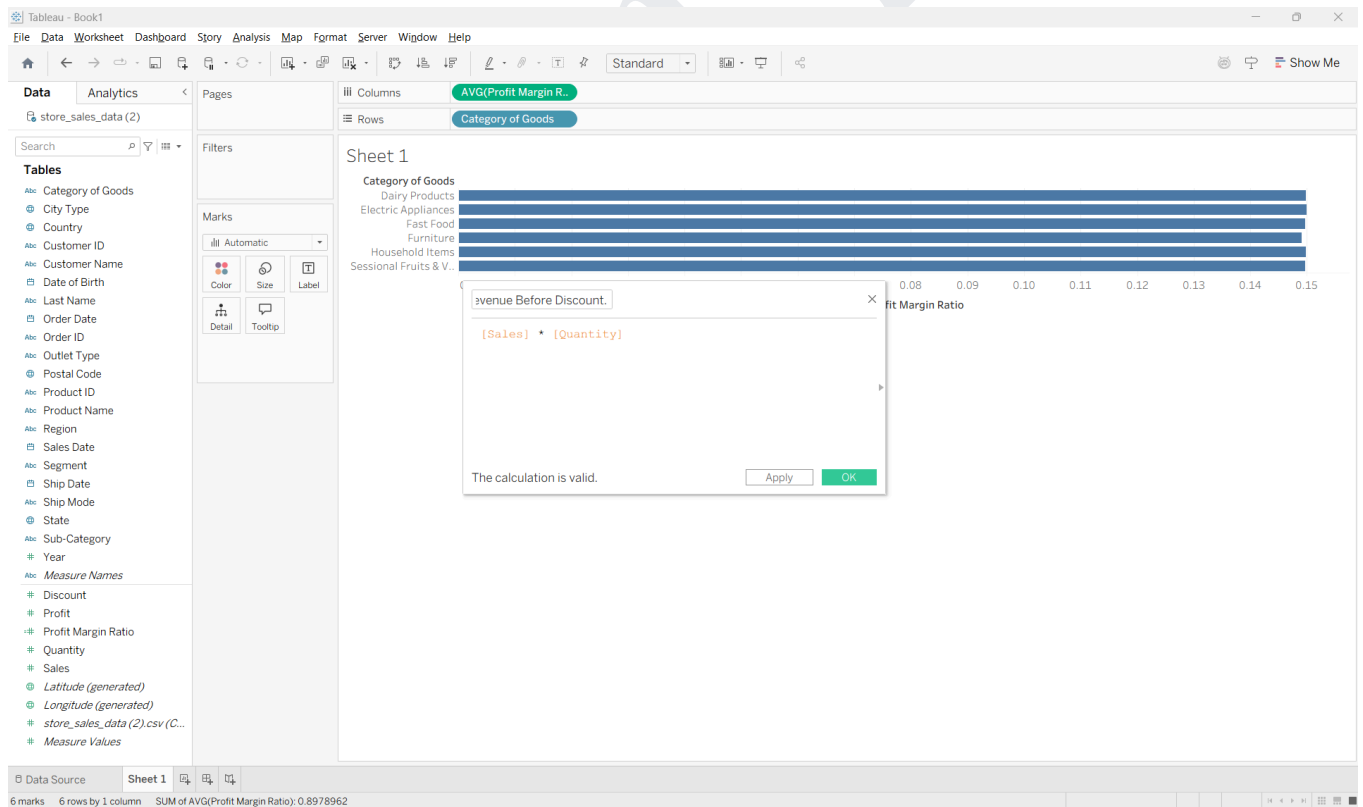


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Name the field: Total Revenue Before Discount.

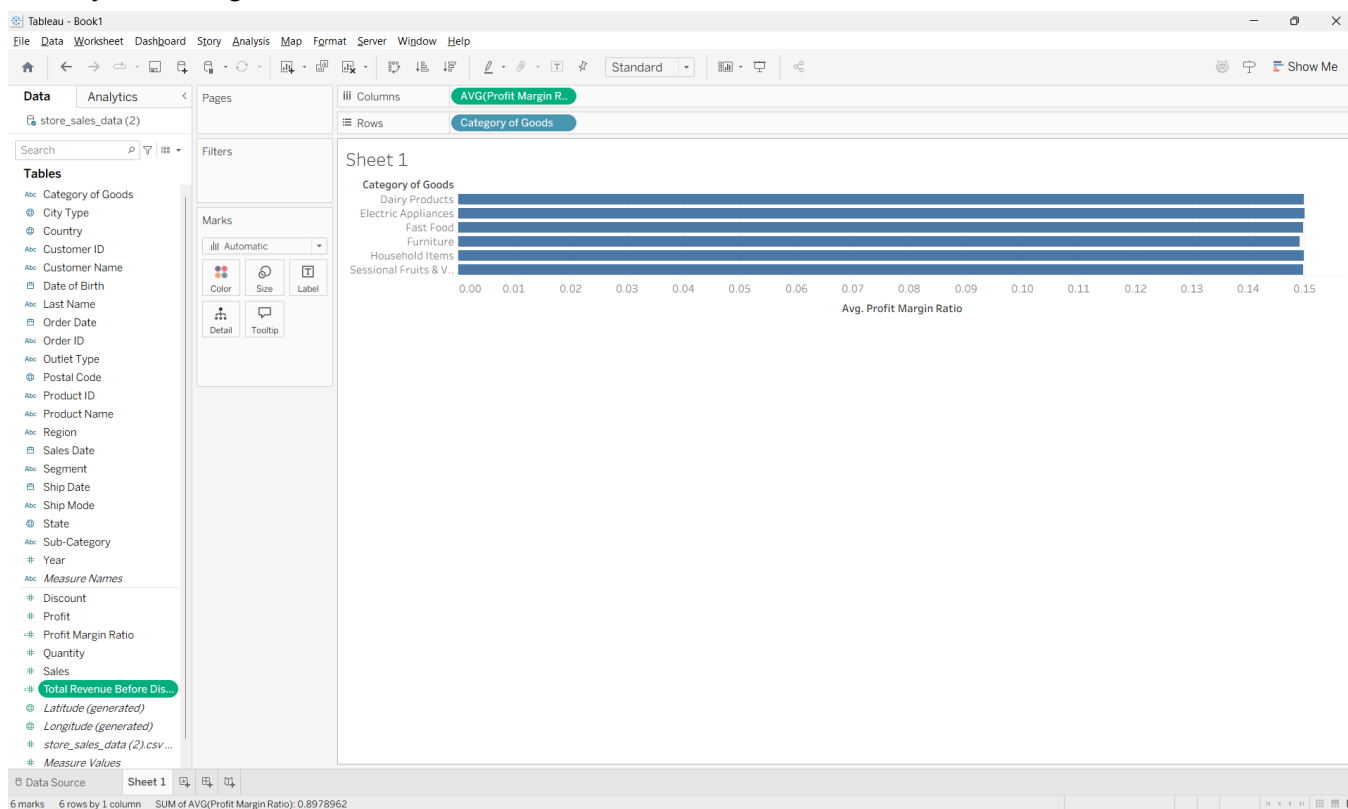


3. In the calculation formula canvas area, input the following expression:
[Sales] * [Quantity]



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

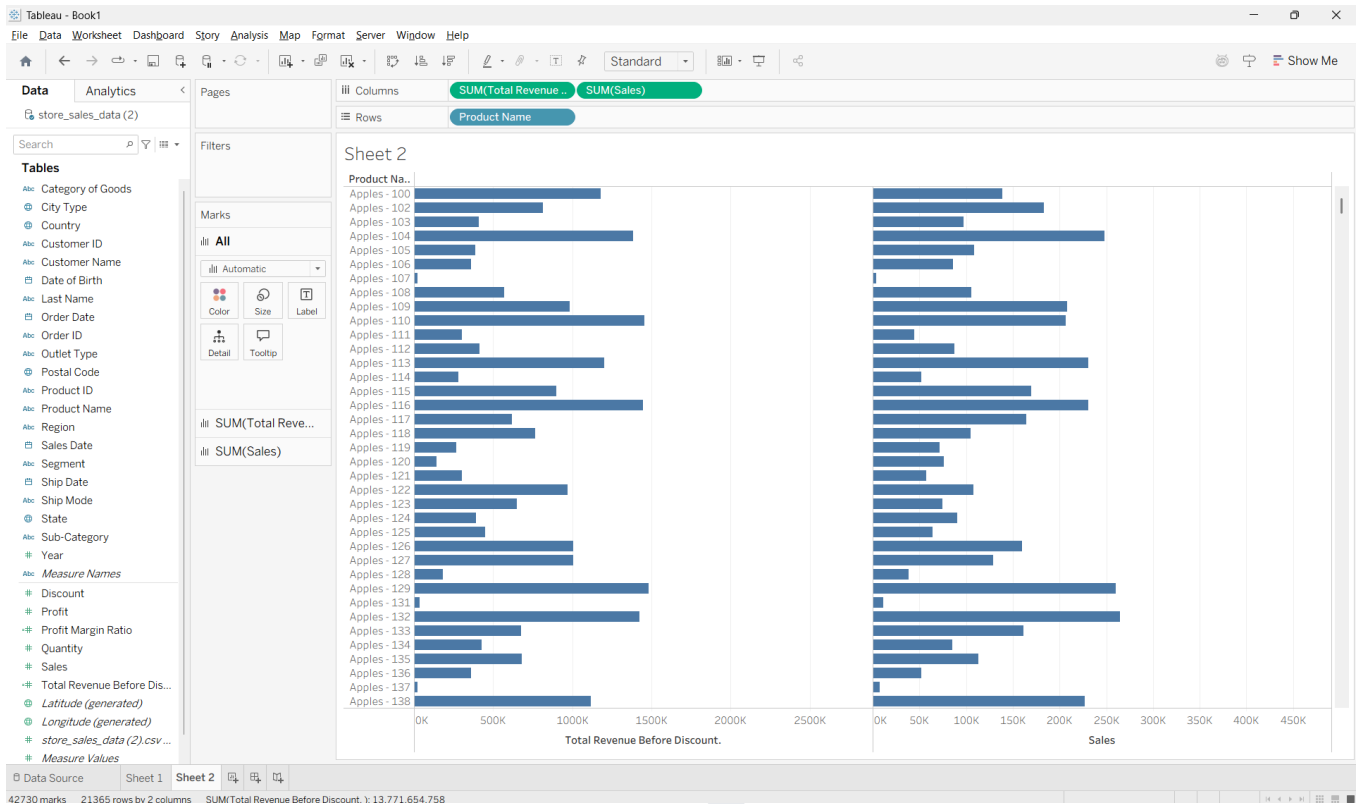
4. Verify the message at the bottom reads "The calculation is valid" and click OK.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

Putting the Field to Use

1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

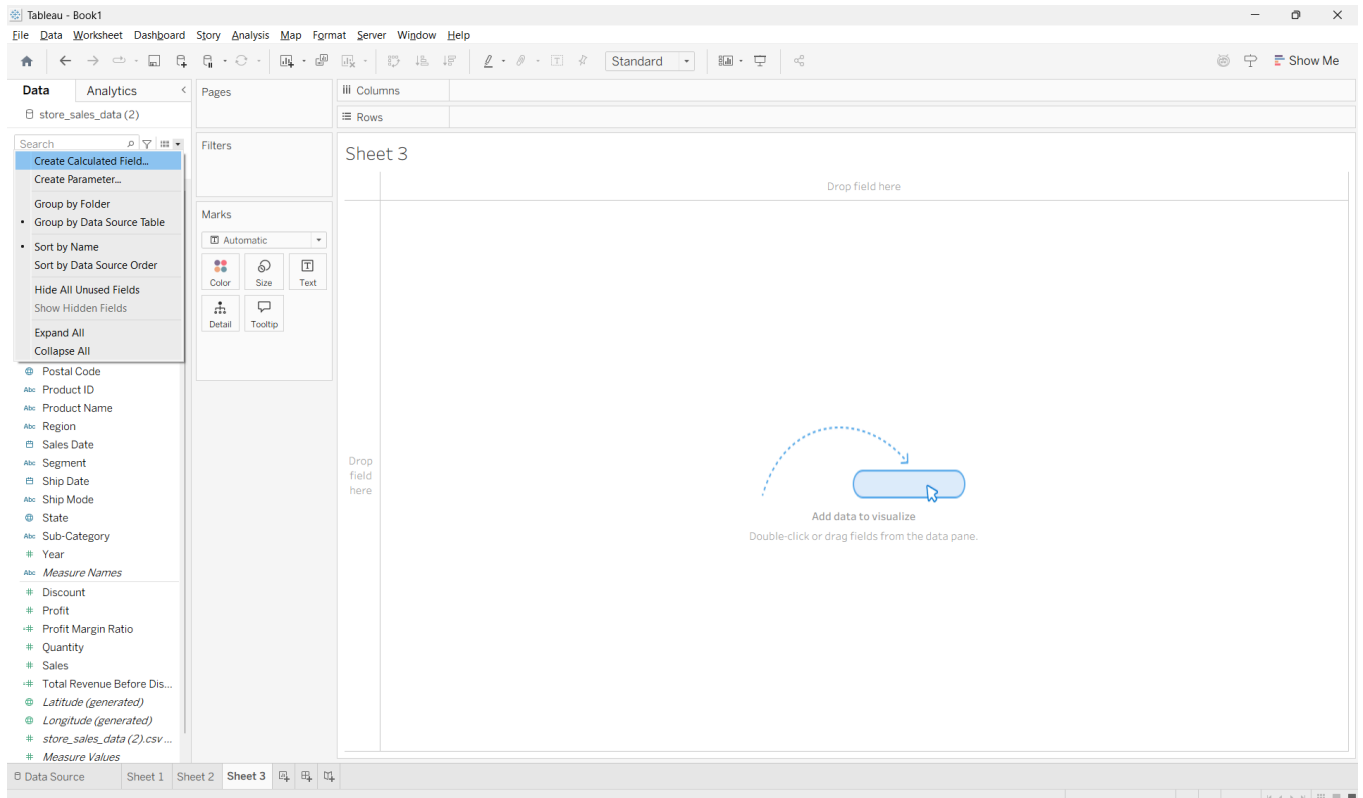


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

C: Performing Aggregate-Level Calculations

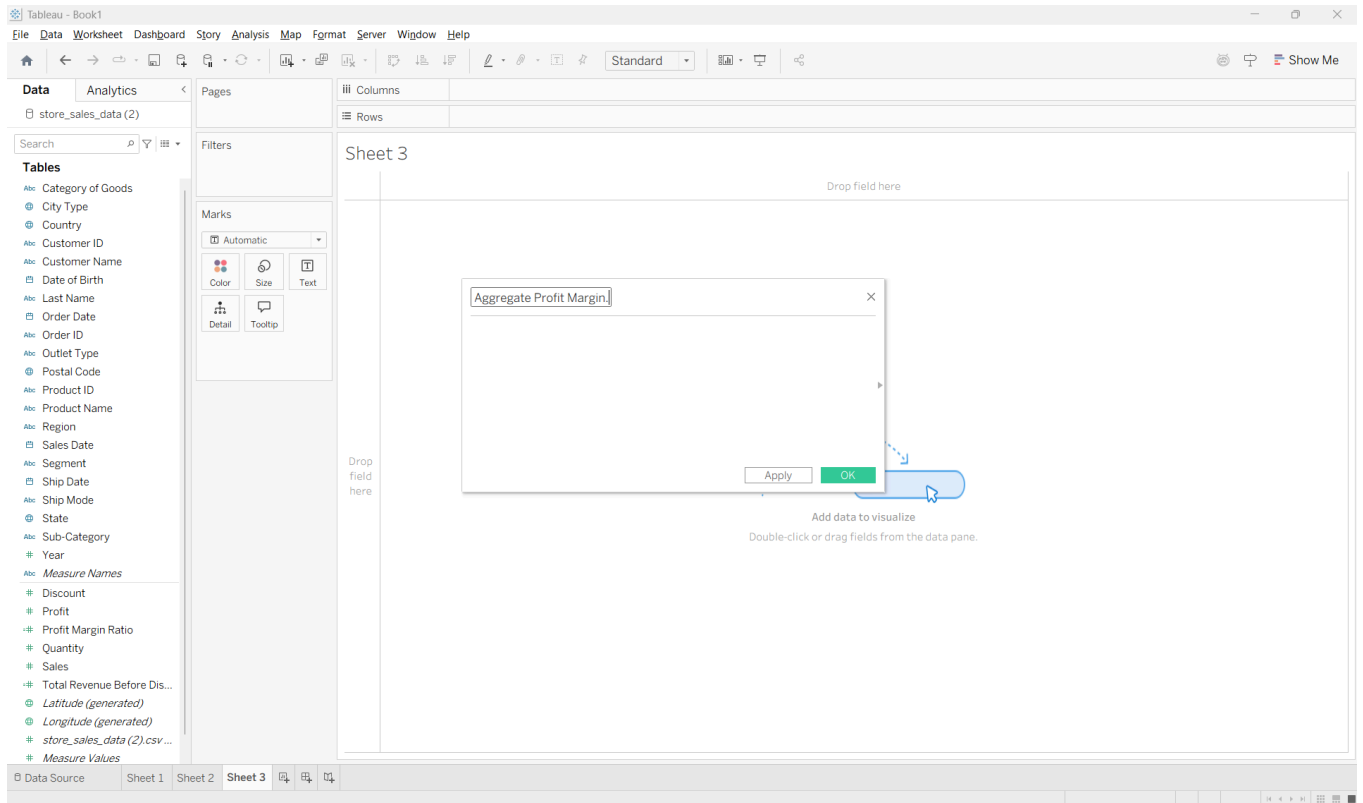
1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

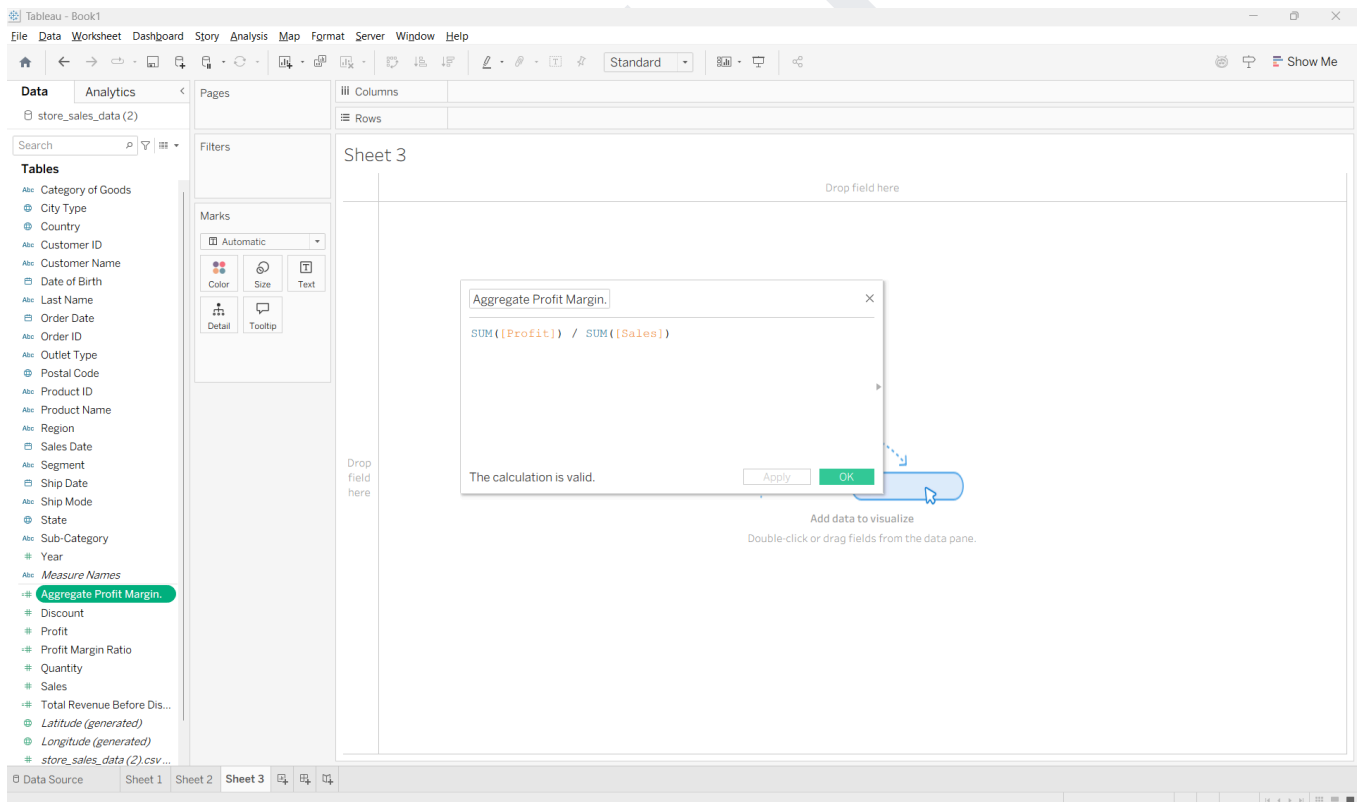


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Name the field: Aggregate Profit Margin.

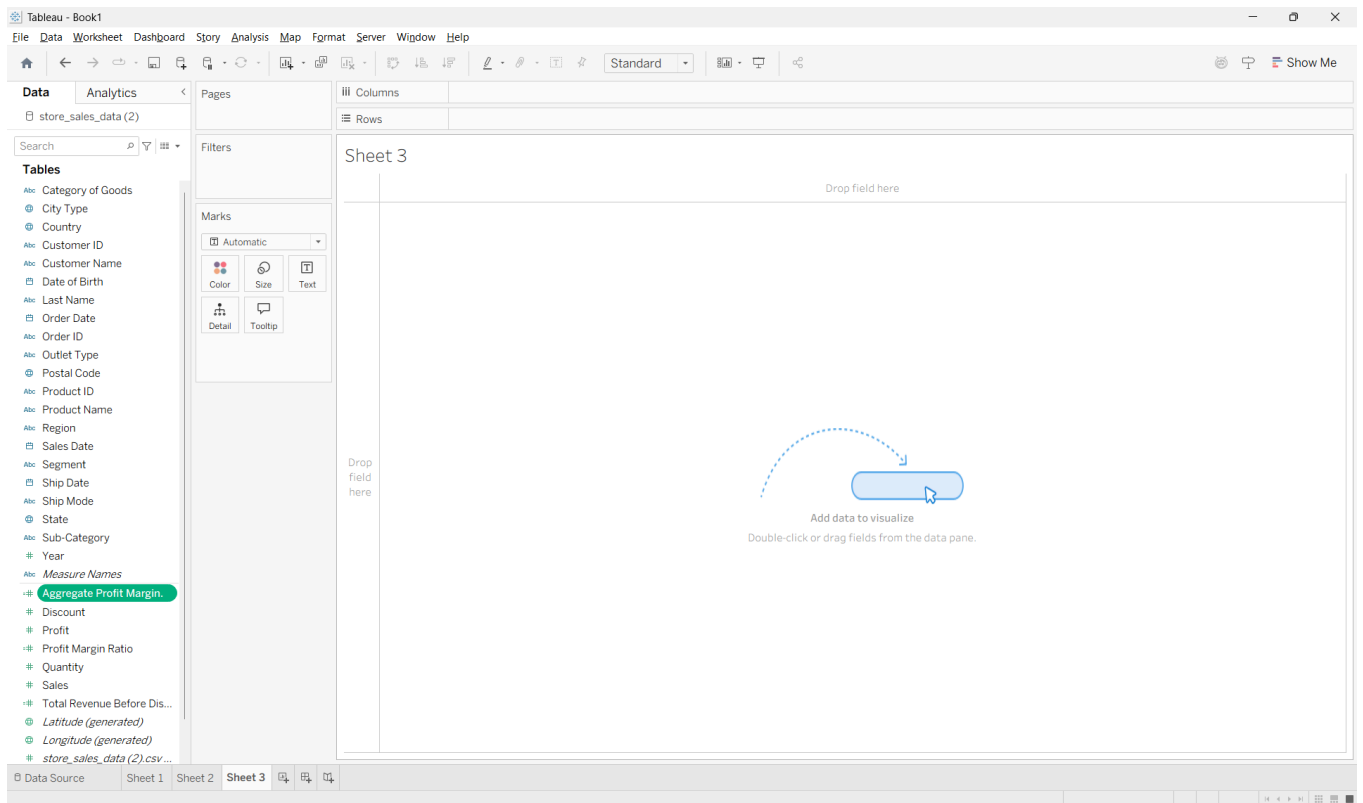


3. In the calculation formula window, type the following expression: $SUM([Profit]) / SUM([Sales])$



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

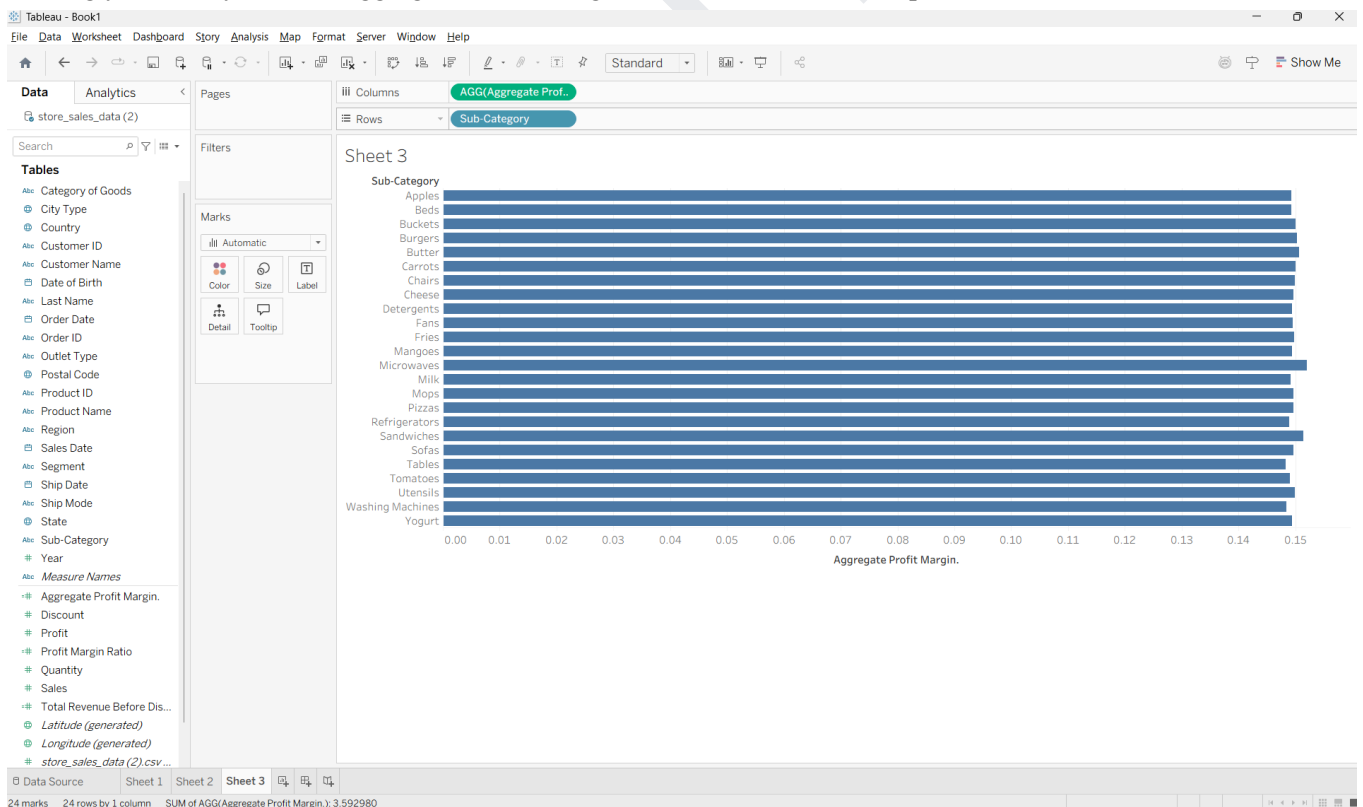
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



2. Putting the Field to Use

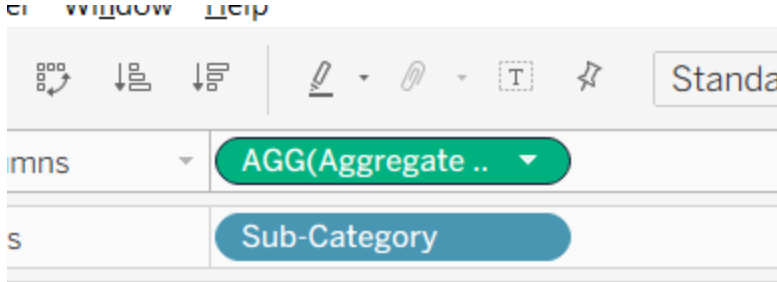
1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.

2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.

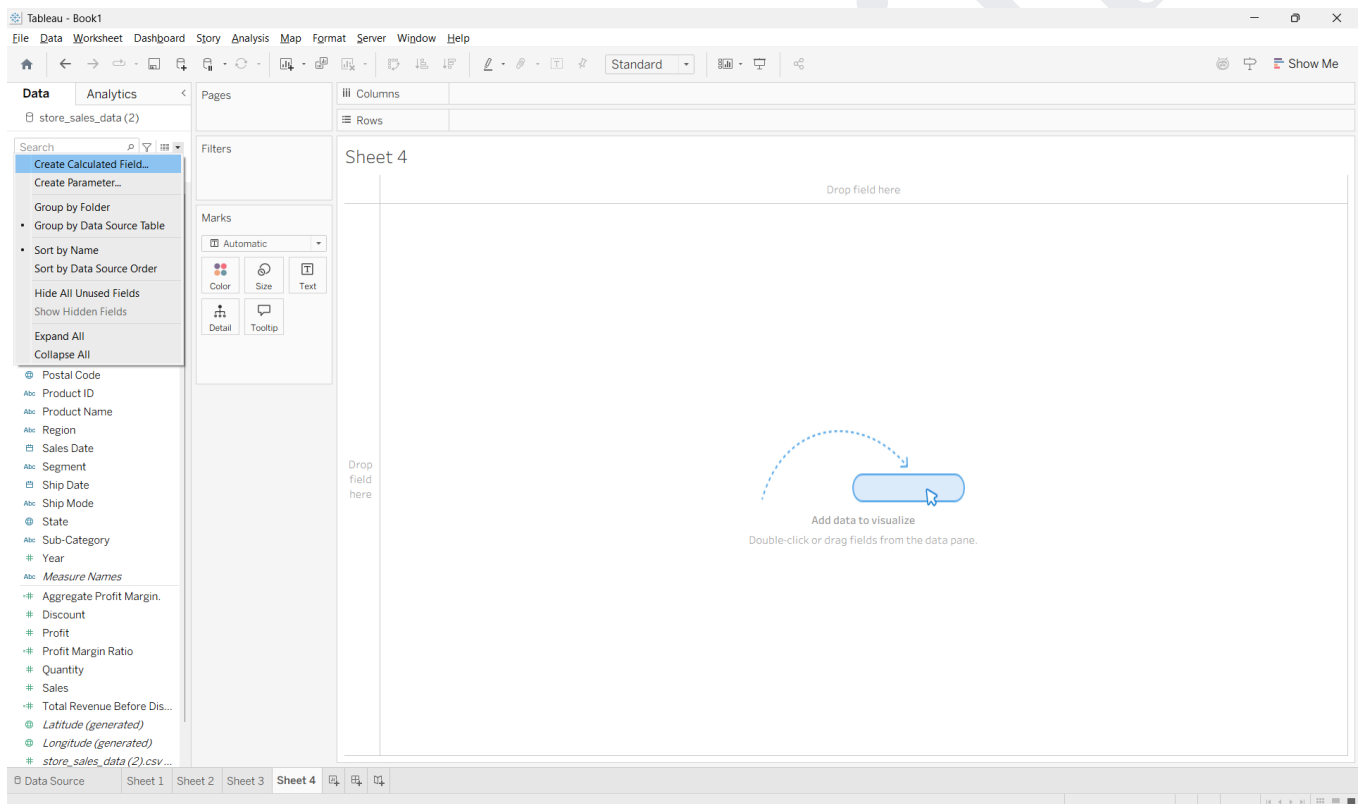


et 3

D: Implementing FIXED Level of Detail (LOD) Calculations

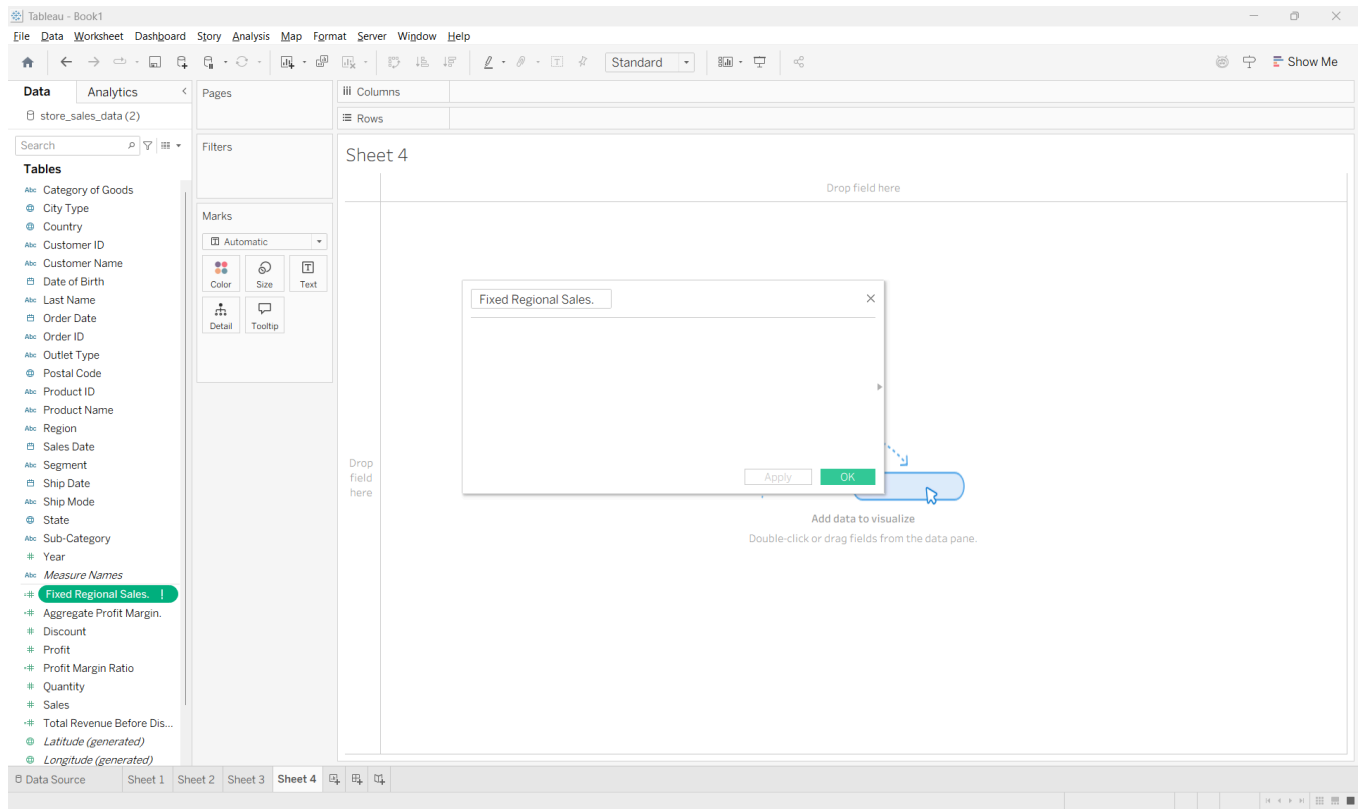
1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

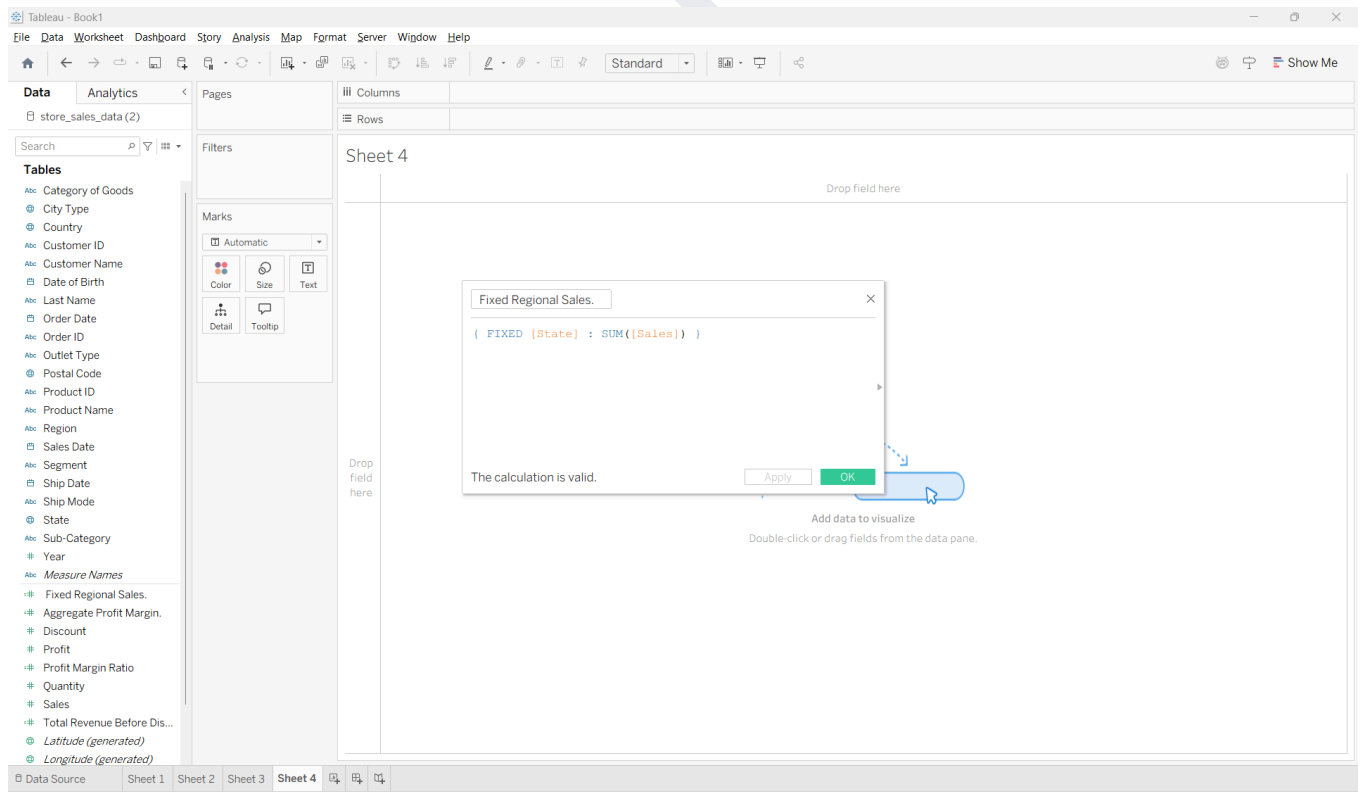
2. Name the field: Fixed Regional Sales.



3. In the calculation formula window, type the following expression using curly braces {}:

{ FIXED [State] : SUM([Sales]) }

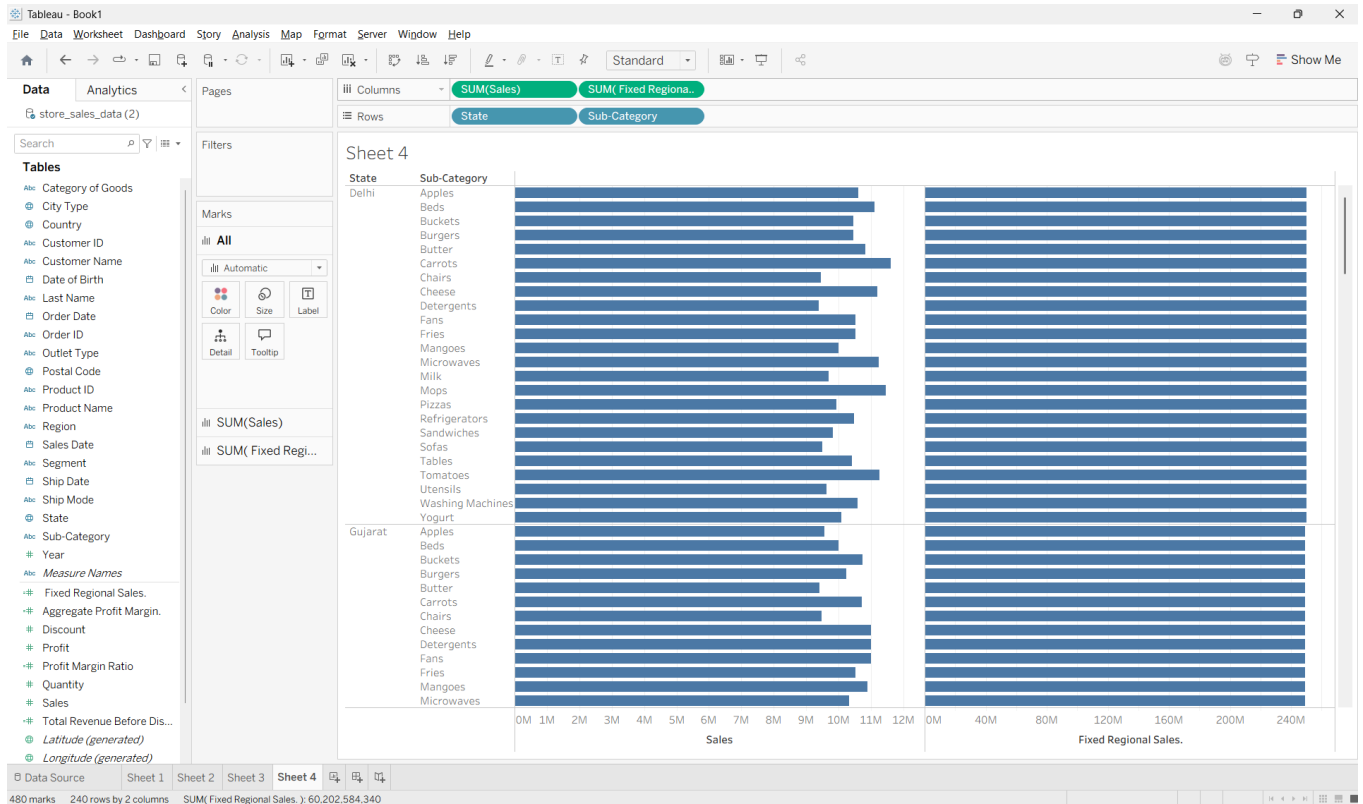
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Putting the Field to Use

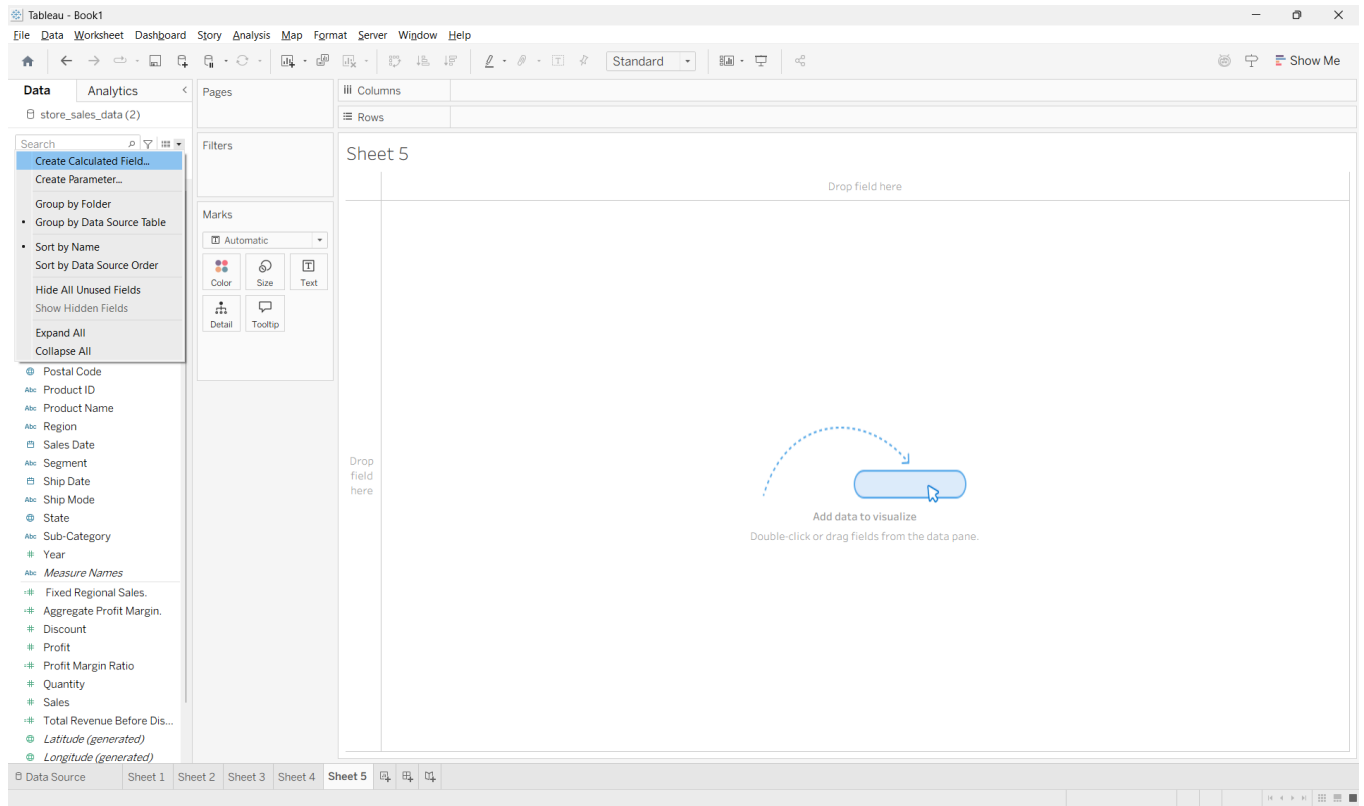
1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf. 2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.



E: Using INCLUDE and EXCLUDE LOD Expressions

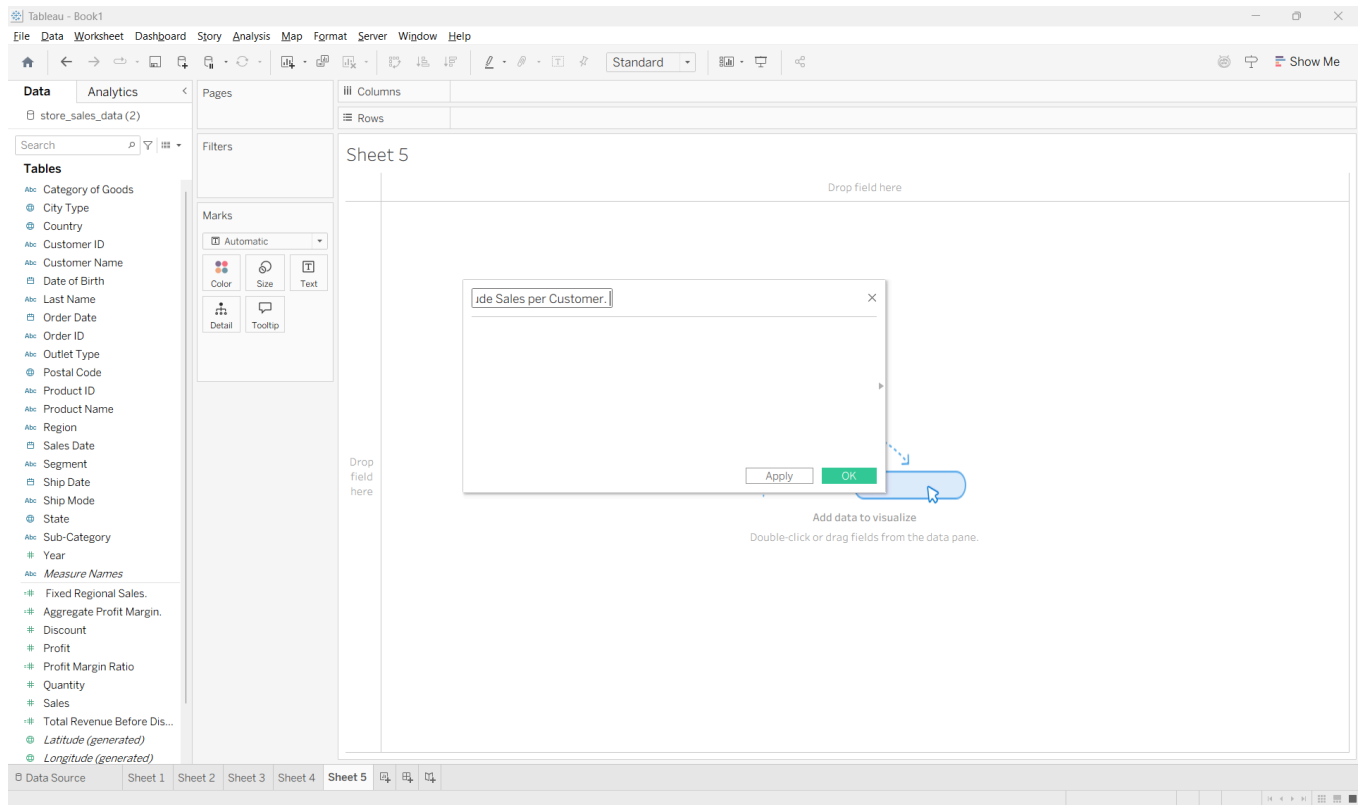
1. Creating an INCLUDE LOD Calculated Field

- Right-click in the blank white space of your Data Pane and select Create Calculated Field...

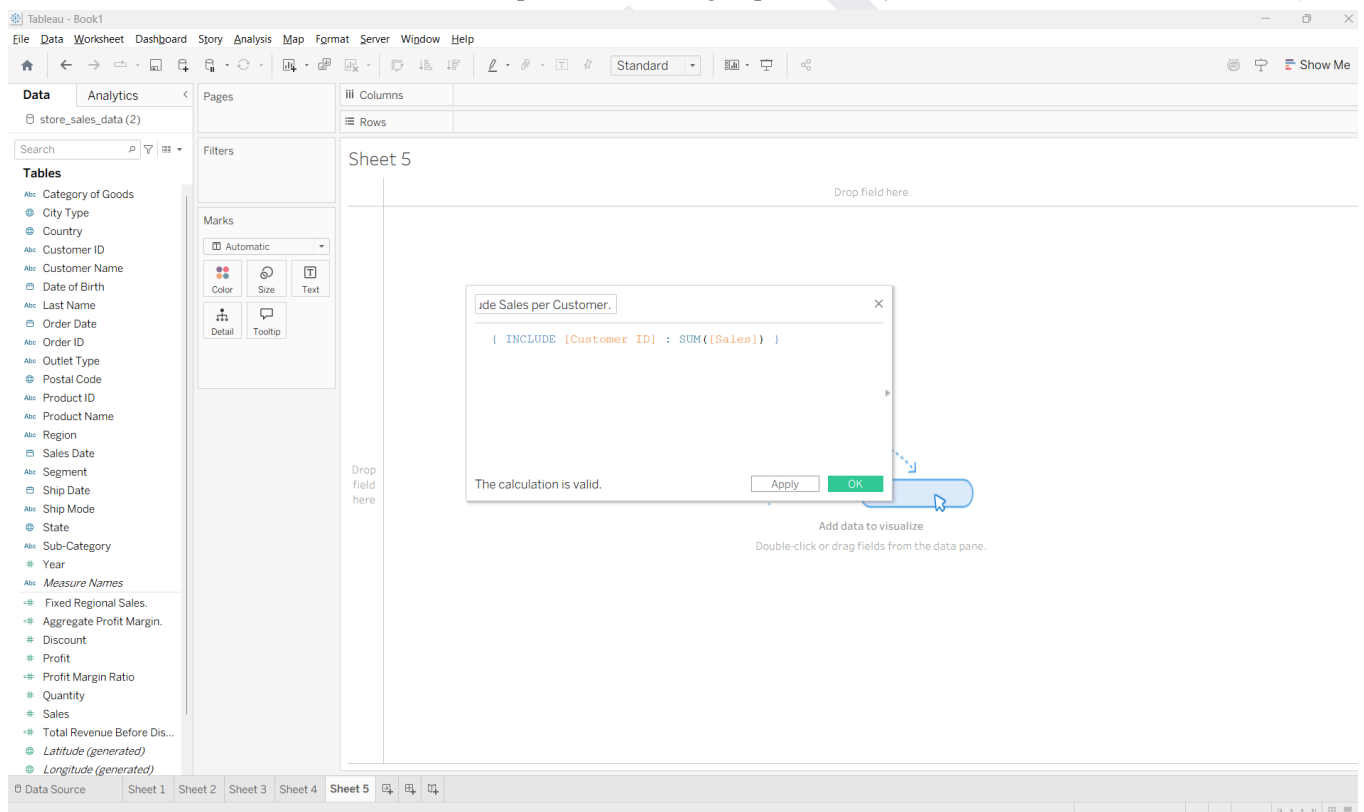


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Name the field: Include Sales per Customer.



3. In the calculation formula canvas area, input the following expression: { INCLUDE [Customer ID] : SUM([Sales]) }

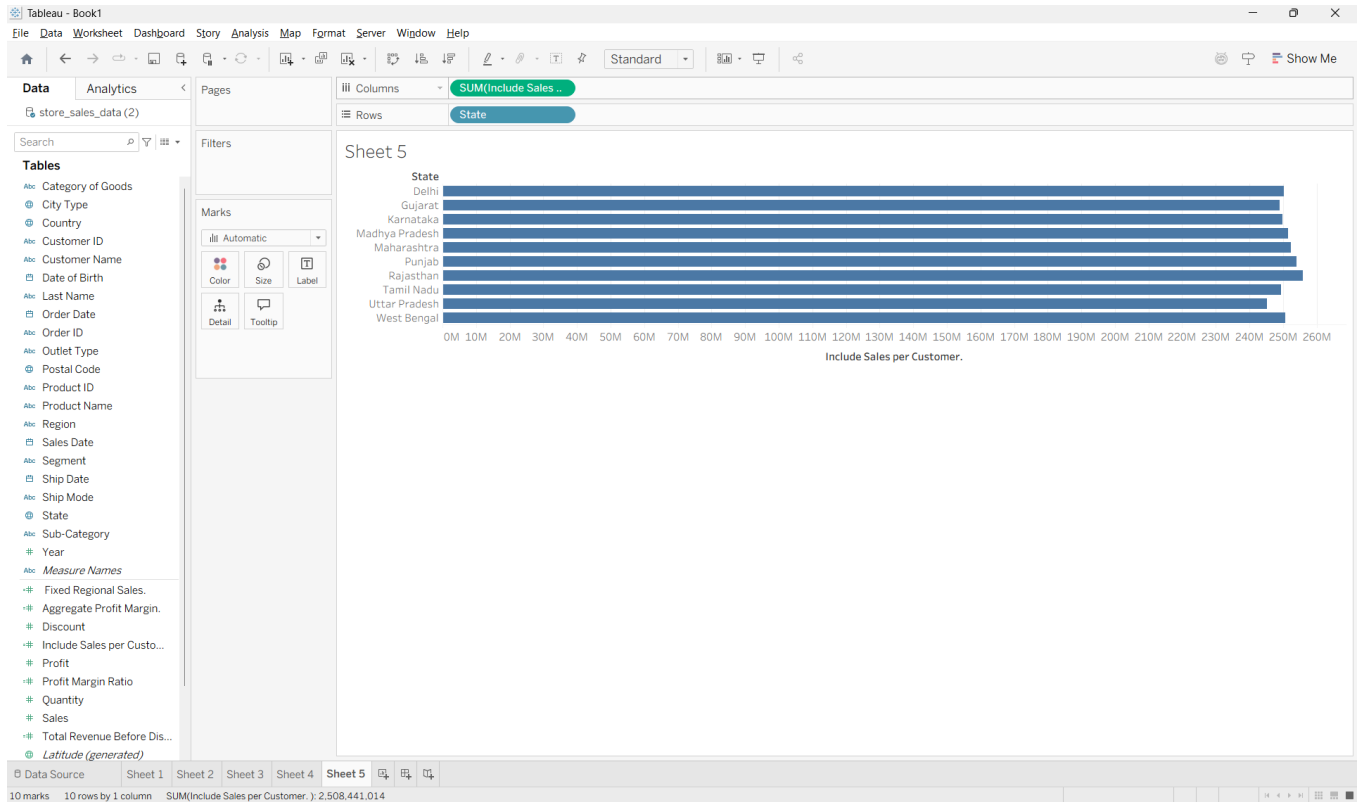


4. Click OK.

Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

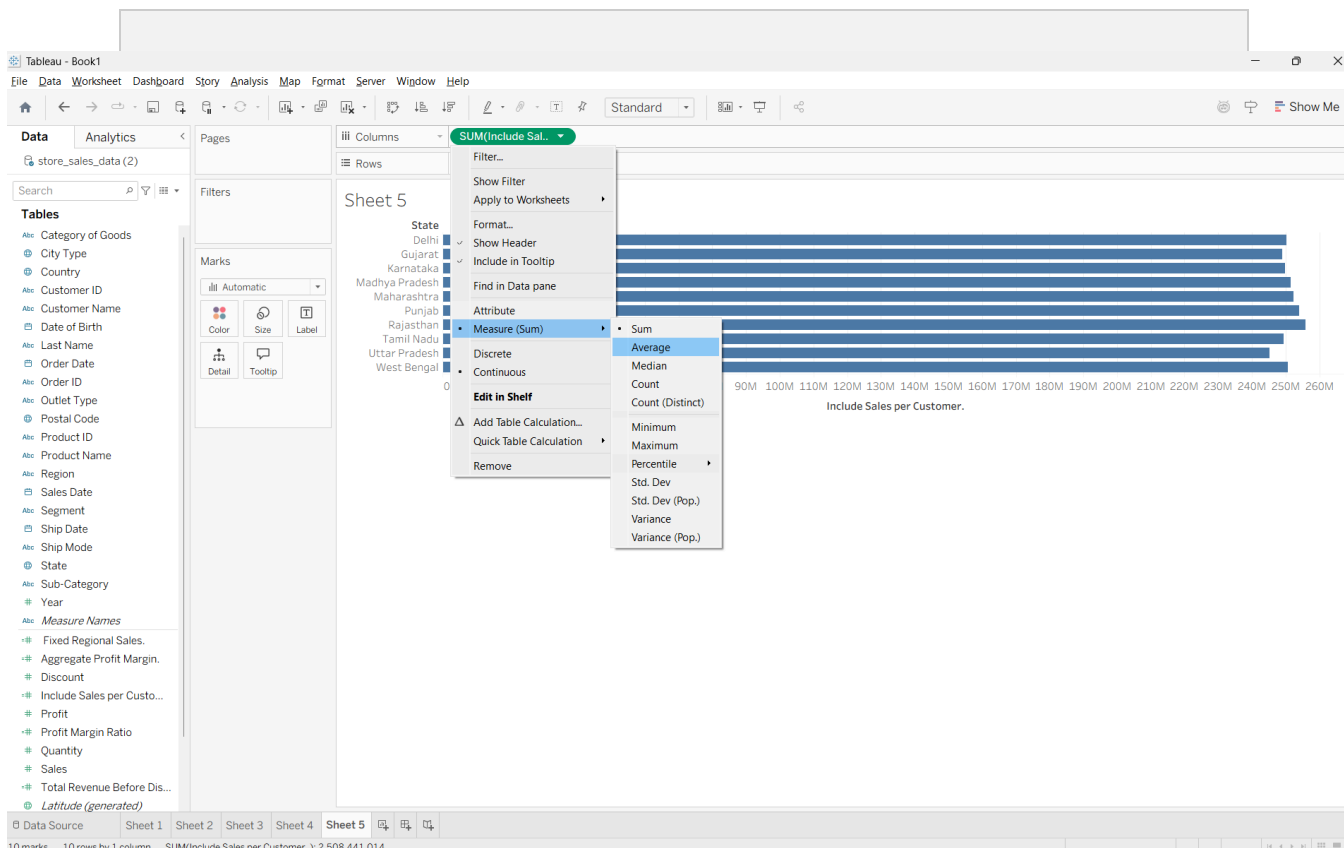
2. Putting the INCLUDE Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Creating an EXCLUDE LOD Calculated Field

1. Open a new blank worksheet.
2. Drag Category of Goods onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).

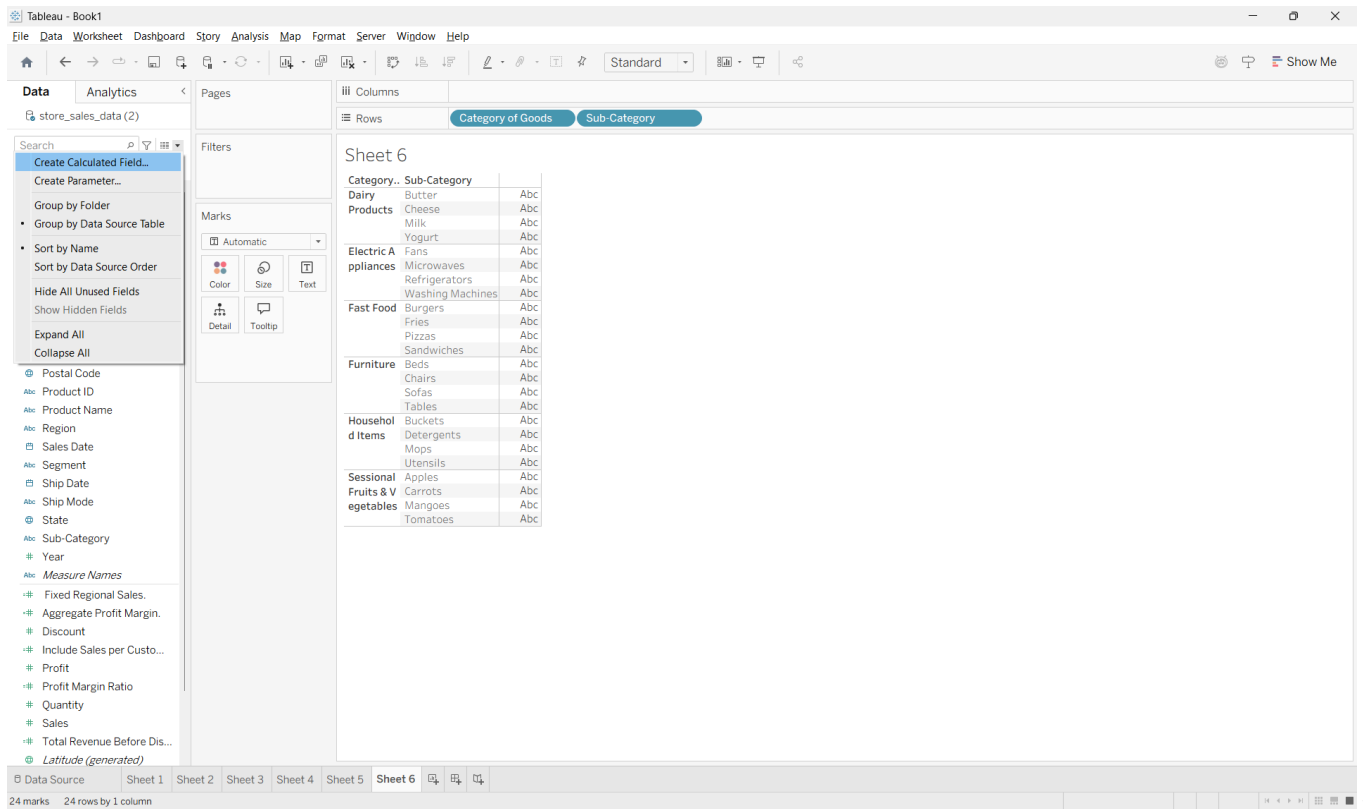
The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Category of Goods' and 'Sub-Category'. The 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic'. The 'Data Source' pane on the left shows a list of fields including 'Category of Goods', 'City Type', 'Country', 'Customer ID', 'Customer Name', 'Date of Birth', 'Last Name', 'Order Date', 'Order ID', 'Outlet Type', 'Postal Code', 'Product ID', 'Product Name', 'Region', 'Sales Date', 'Segment', 'Ship Date', 'Ship Mode', 'State', 'Sub-Category', 'Year', 'Measure Names', 'Fixed Regional Sales', 'Aggregate Profit Margin', 'Discount', 'Include Sales per Custo...', 'Profit', 'Profit Margin Ratio', 'Quantity', 'Sales', 'Total Revenue Before Dis...', and 'Latitude (generated)'. The 'Sheet 6' view displays a table with the following data:

Category..	Sub-Category	
Dairy	Butter	Abc
Products	Cheese	Abc
	Milk	Abc
	Yogurt	Abc
Electric Appliances	Fans	Abc
	Microwaves	Abc
	Refrigerators	Abc
	Washing Machines	Abc
Fast Food	Burgers	Abc
	Fries	Abc
	Pizzas	Abc
	Sandwiches	Abc
Furniture	Beds	Abc
	Chairs	Abc
	Sofas	Abc
	Tables	Abc
Household Items	Buckets	Abc
	Detergents	Abc
	Mops	Abc
	Utensils	Abc
Sessional Fruits & Vegetables	Apples	Abc
	Carrots	Abc
	Mangoes	Abc
	Tomatoes	Abc

Sheth L.U.J & Sir M.V College

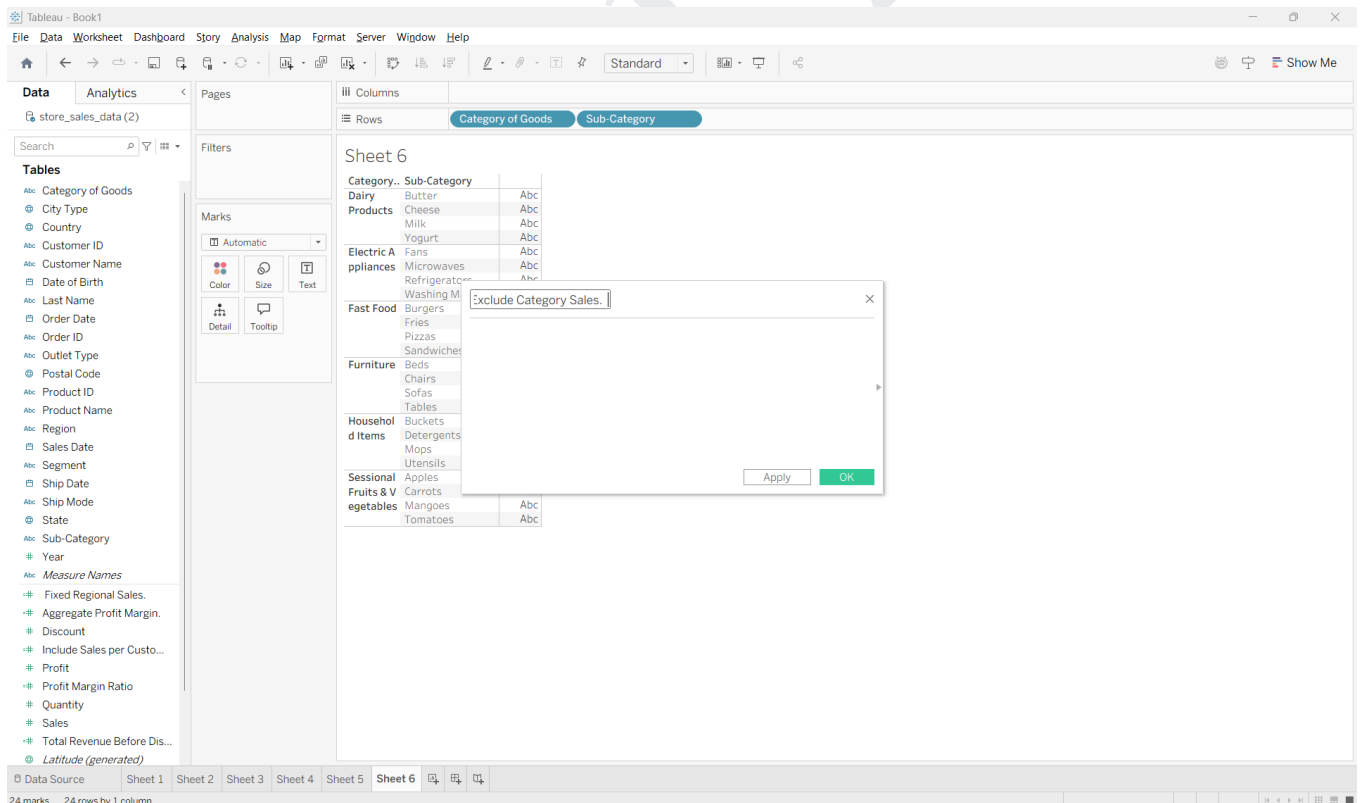
Subject: Data Visualization with Power BI and Tableau

4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



5.

Name the field: Exclude Category Sales.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

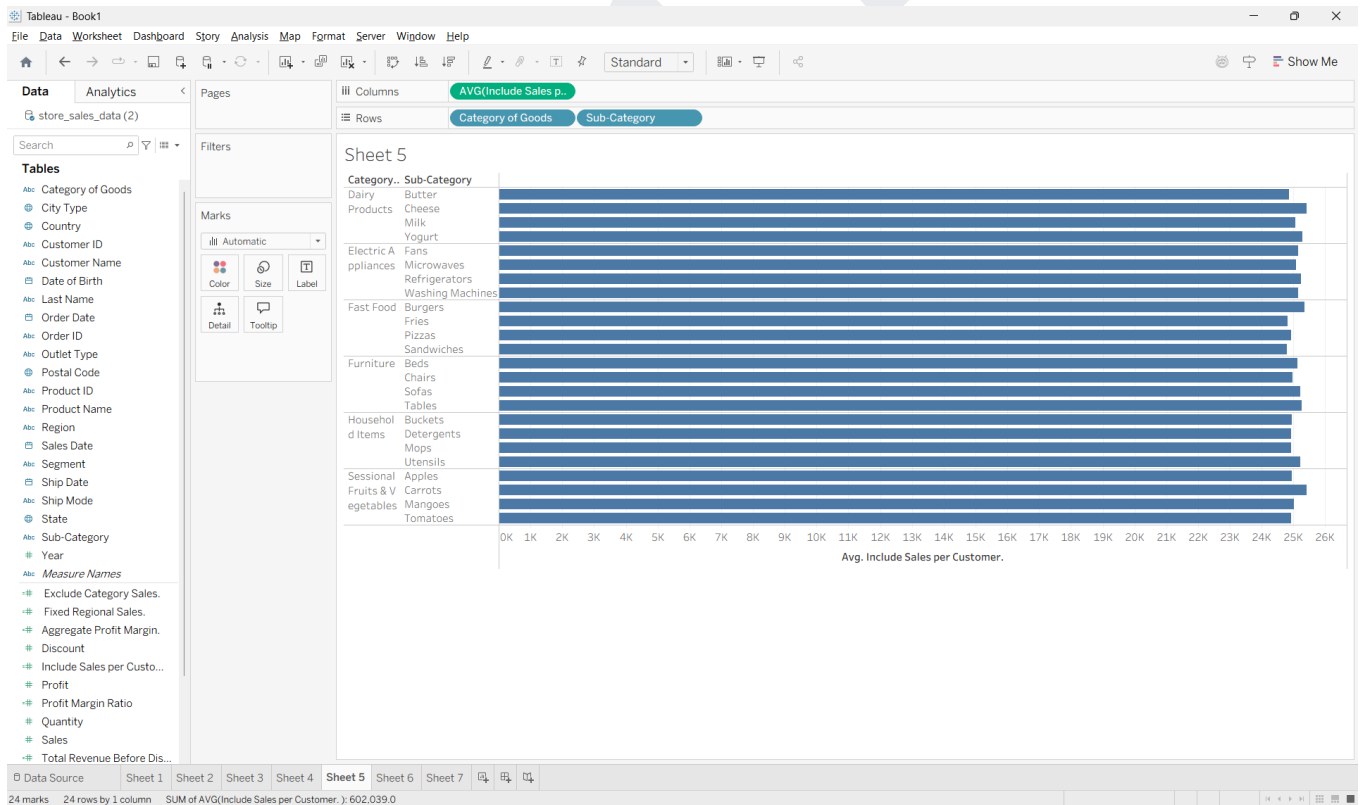
6. Input the following expression:

{ EXCLUDE [Sub-Category] : SUM([Sales]) }

The screenshot shows the Tableau interface with the 'Columns' shelf containing 'Category of Goods' and 'Sub-Category'. A dialog box titled 'Exclude Category Sales.' is open, displaying the calculation: `{ EXCLUDE [Sub-Category] : SUM([Sales]) }`. The dialog also shows a preview of the data table with columns 'Category..', 'Sub-Category', and 'Sales'. The preview data is as follows:

Category..	Sub-Category	Sales
Dairy	Butter	Abc
Dairy	Cheese	Abc
Dairy	Milk	Abc
Dairy	Yogurt	Abc
Electric Appliances	Fans	Abc
Electric Appliances	Microwaves	Abc
Electric Appliances	Refrigerators	Abc
Electric Appliances	Washing Machines	Abc
Fast Food	Burgers	Abc
Fast Food	Fries	Abc
Fast Food	Pizzas	Abc
Fast Food	Sandwiches	Abc
Furniture	Beds	Abc
Furniture	Chairs	Abc
Furniture	Sofas	Abc
Furniture	Tables	Abc
Household Items	Buckets	Abc
Household Items	Detergents	Abc
Household Items	Mops	Abc
Household Items	Utensils	Abc
Seasonal Fruits & Vegetables	Apples	Abc
Seasonal Fruits & Vegetables	Carrots	Abc
Seasonal Fruits & Vegetables	Mangoes	Abc
Seasonal Fruits & Vegetables	Tomatoes	Abc

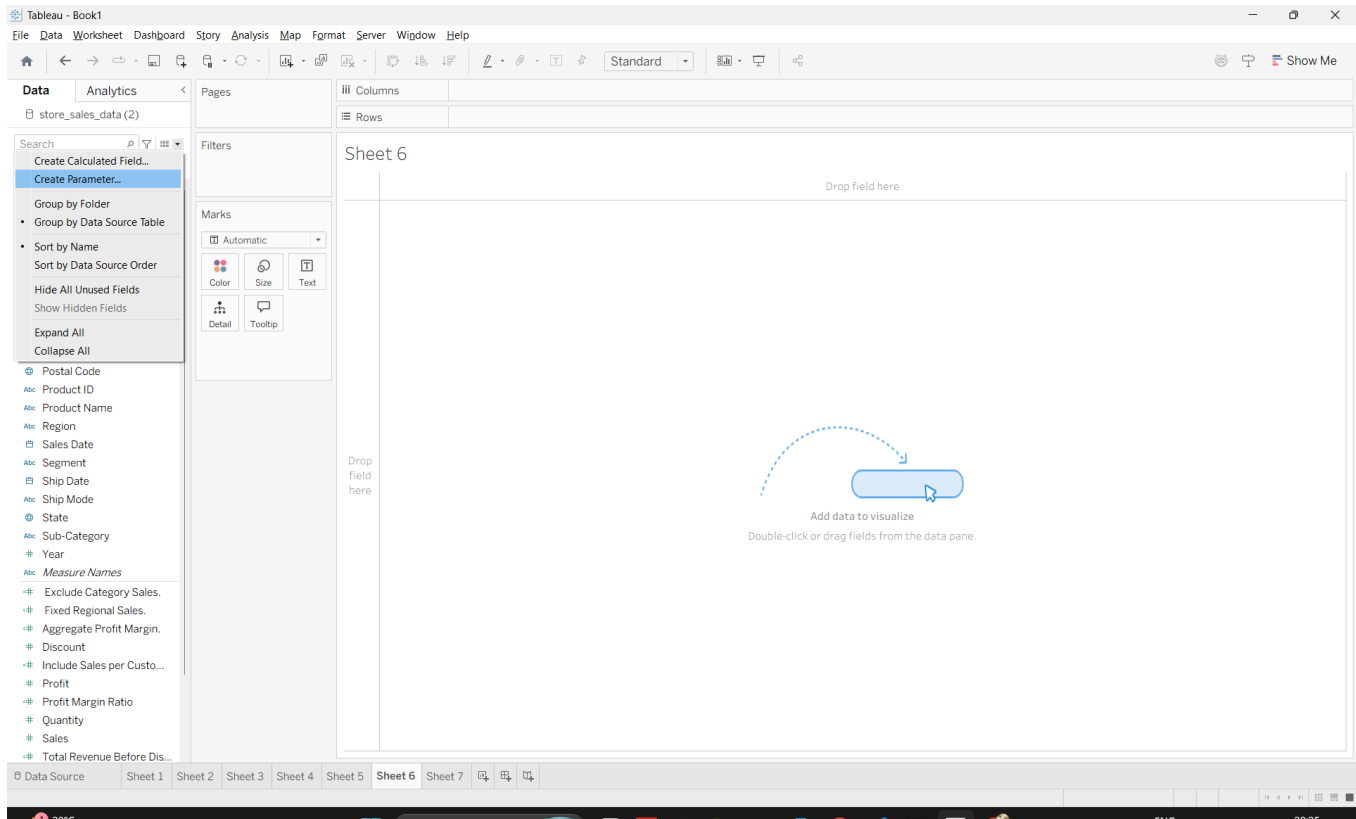
7. Click OK and drag this new field onto the Columns shelf.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

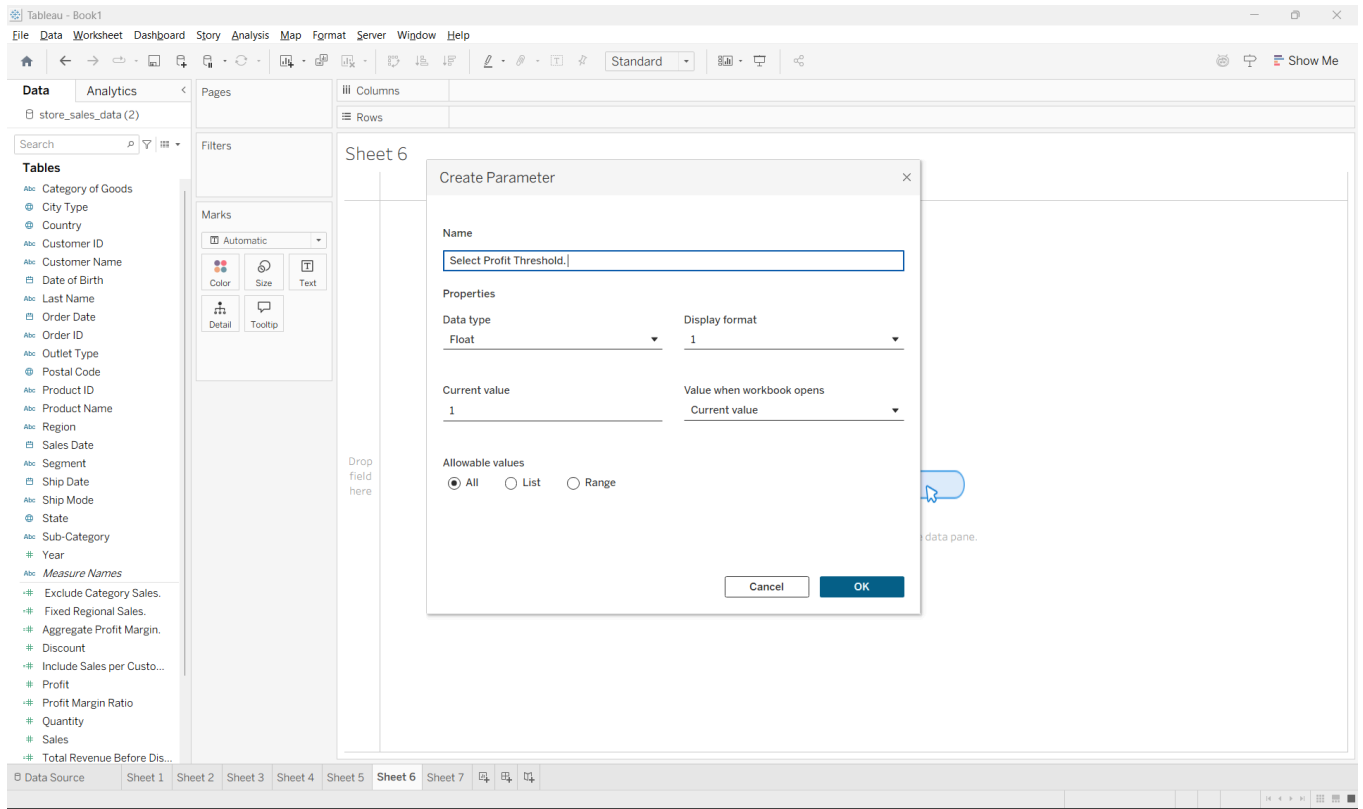
Let's build a parameter that lets users set their own dynamic Profit Target threshold:

1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose

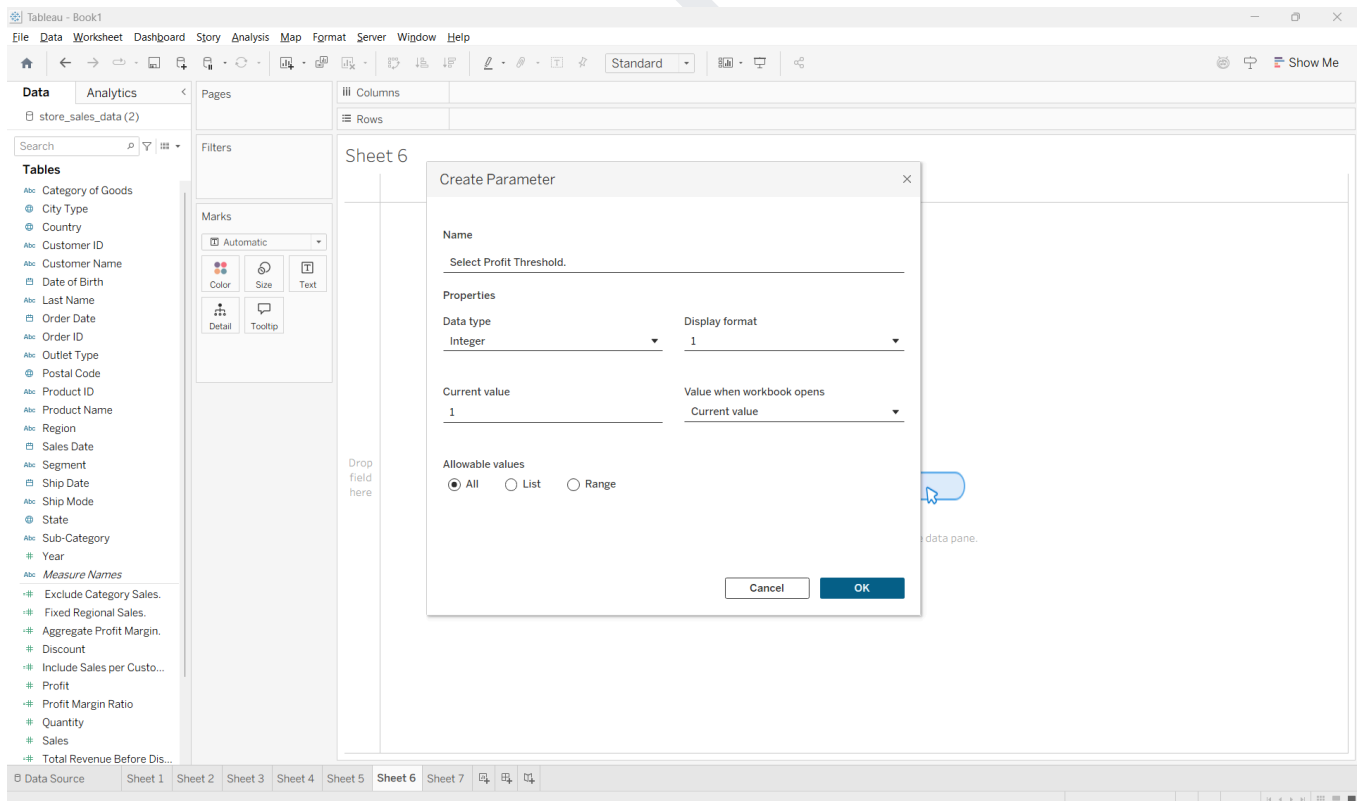


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Name the parameter: Select Profit Threshold.

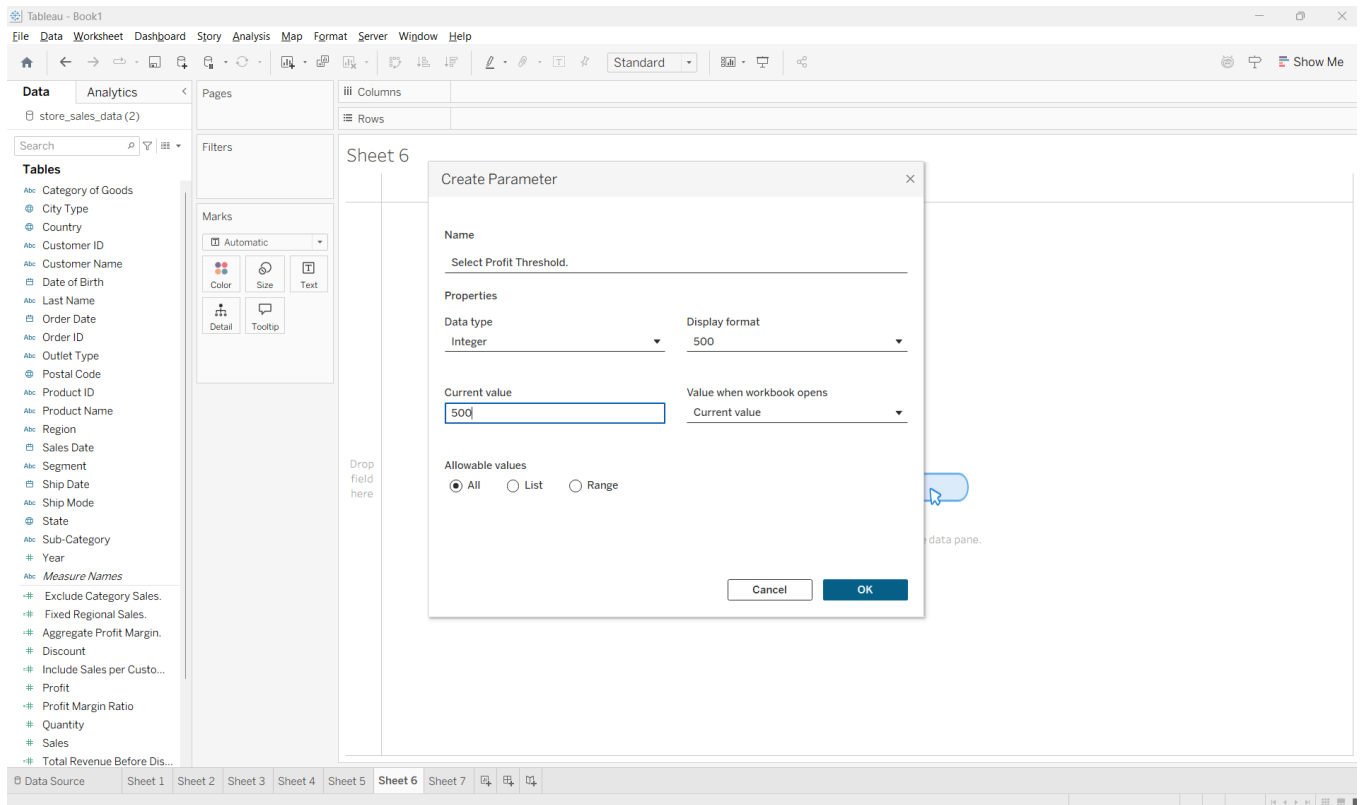


3. Set the Data Type to Float (or Integer).

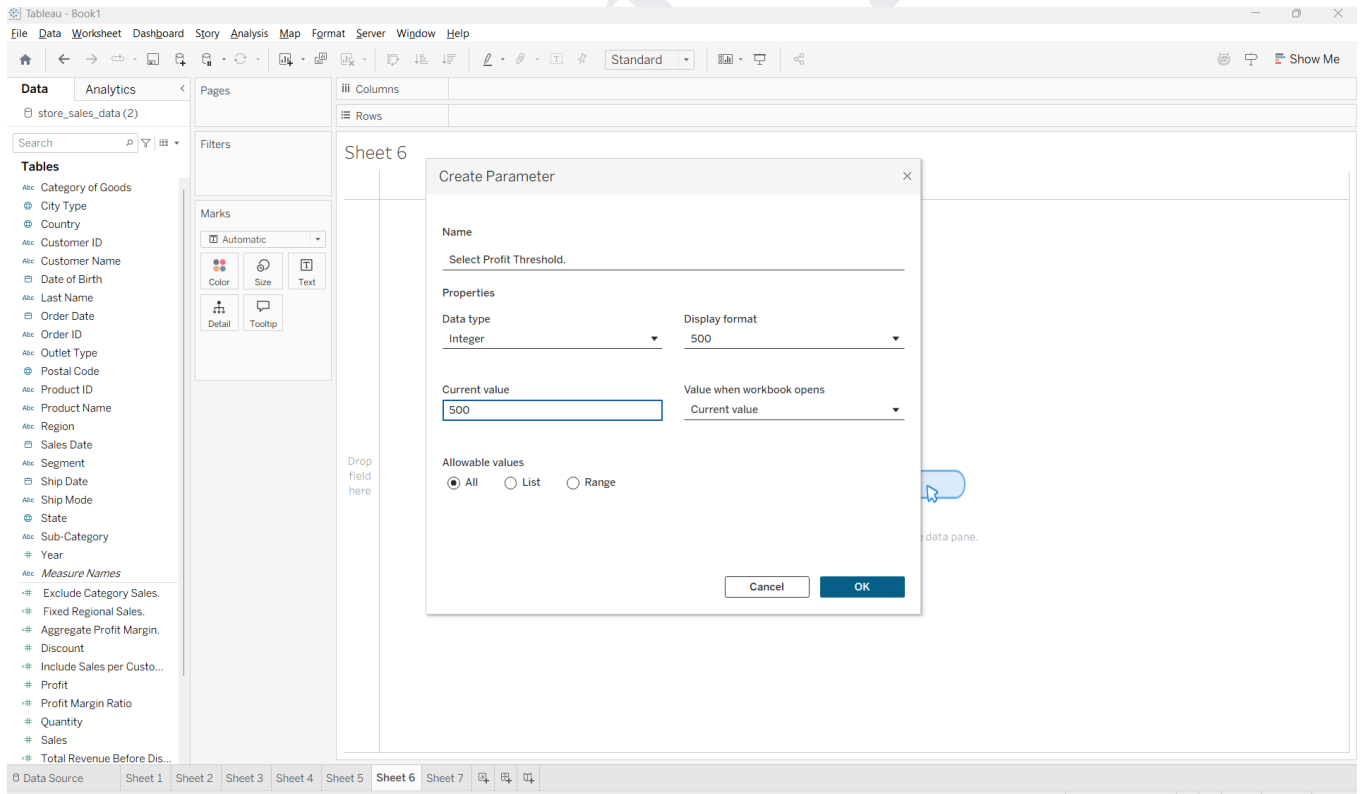


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

4. Set the Current Value to 500.

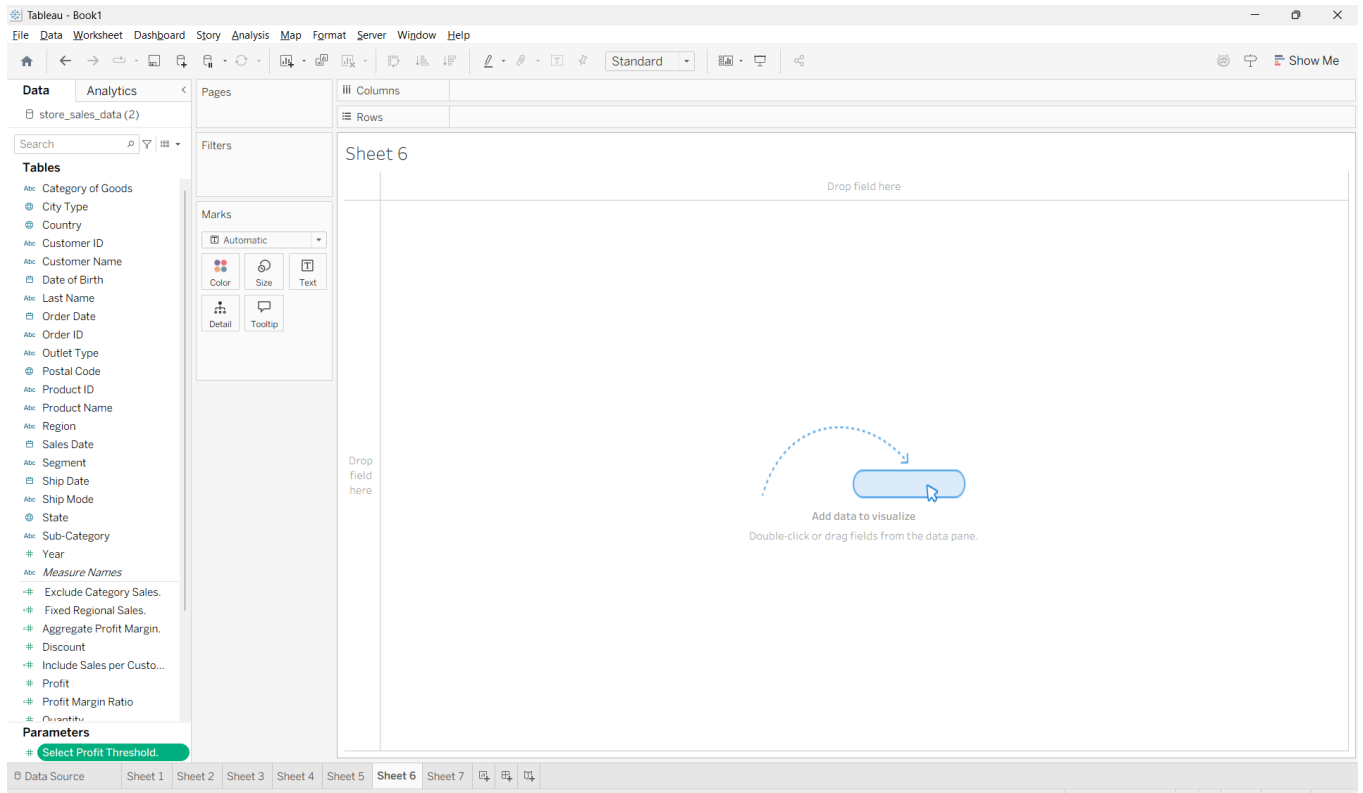


5. Keep the Allowable values set to All and click OK.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

6. Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.



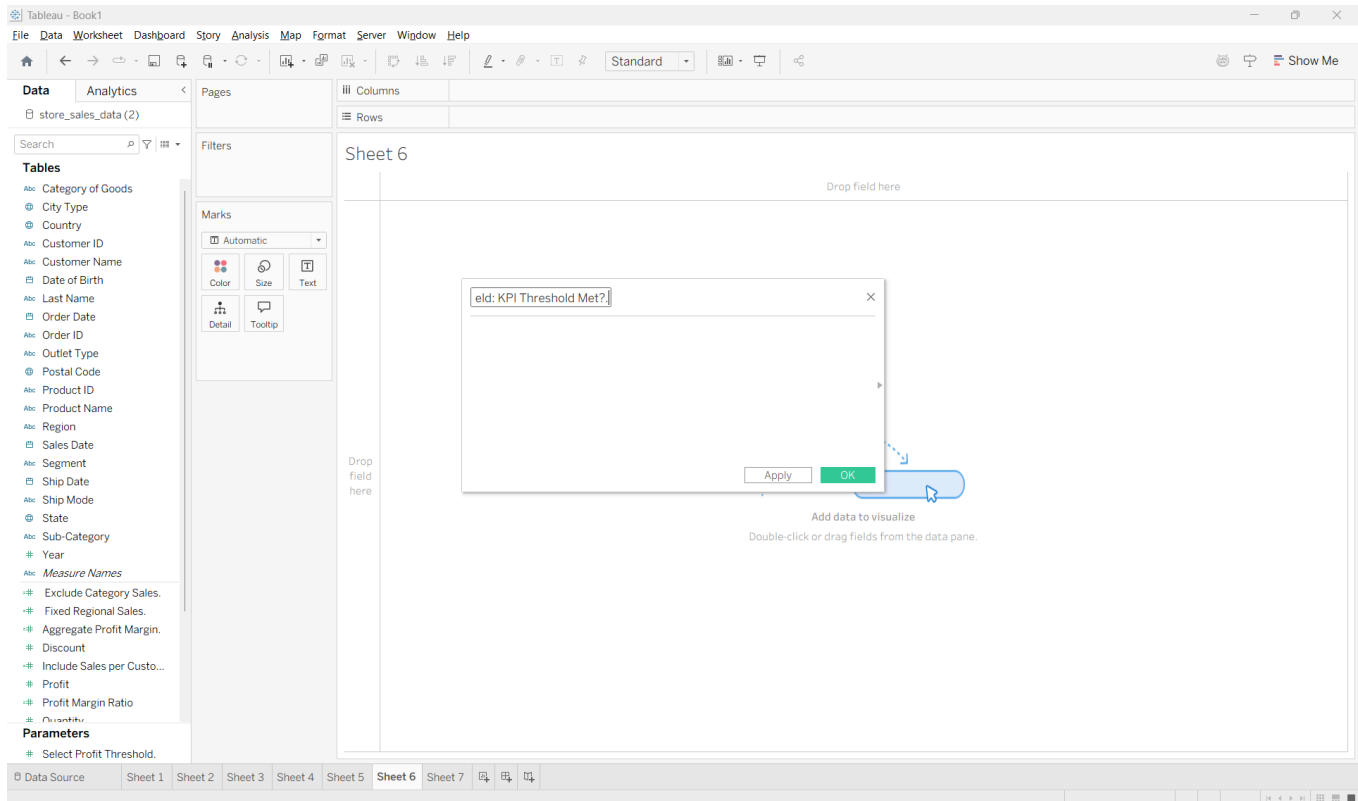
Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...

2. Name this field: KPI Threshold Met?.

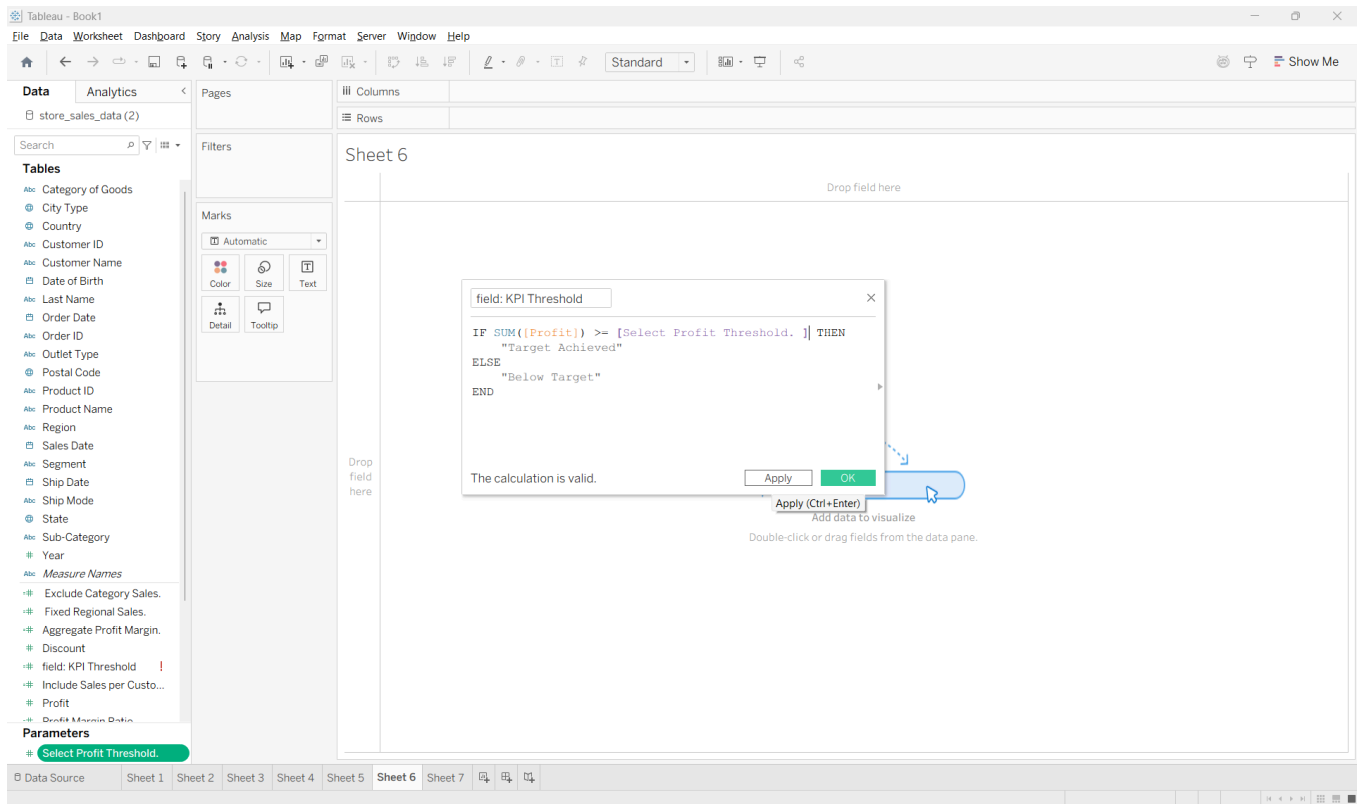


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Input the following logical IF-THEN conditional statement:

IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved" ELSE "Below Target"

END (NOTE: THE SELECT PROFIT THRESHOLD SHOULD TURN INTO PURPLE IF IT IS NOT TURNING IN PURPLE PLEASE RETYPE IT AGAIN)



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

4. Verify the formula is valid and click OK.

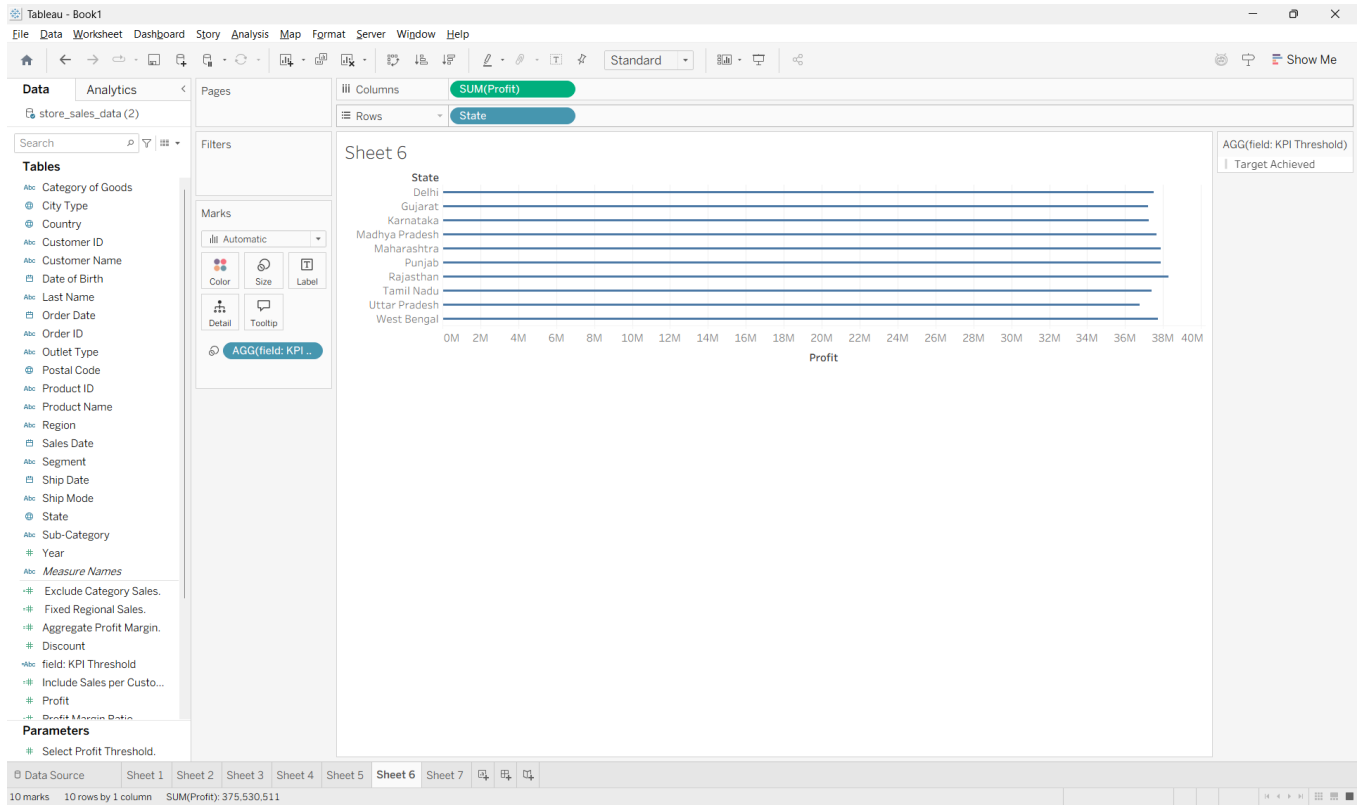
3. Putting the Interactive Parameter to Use

1. Open a new blank worksheet.

2. Drag the State dimension from your Data Pane and drop it onto the Rows shelf.

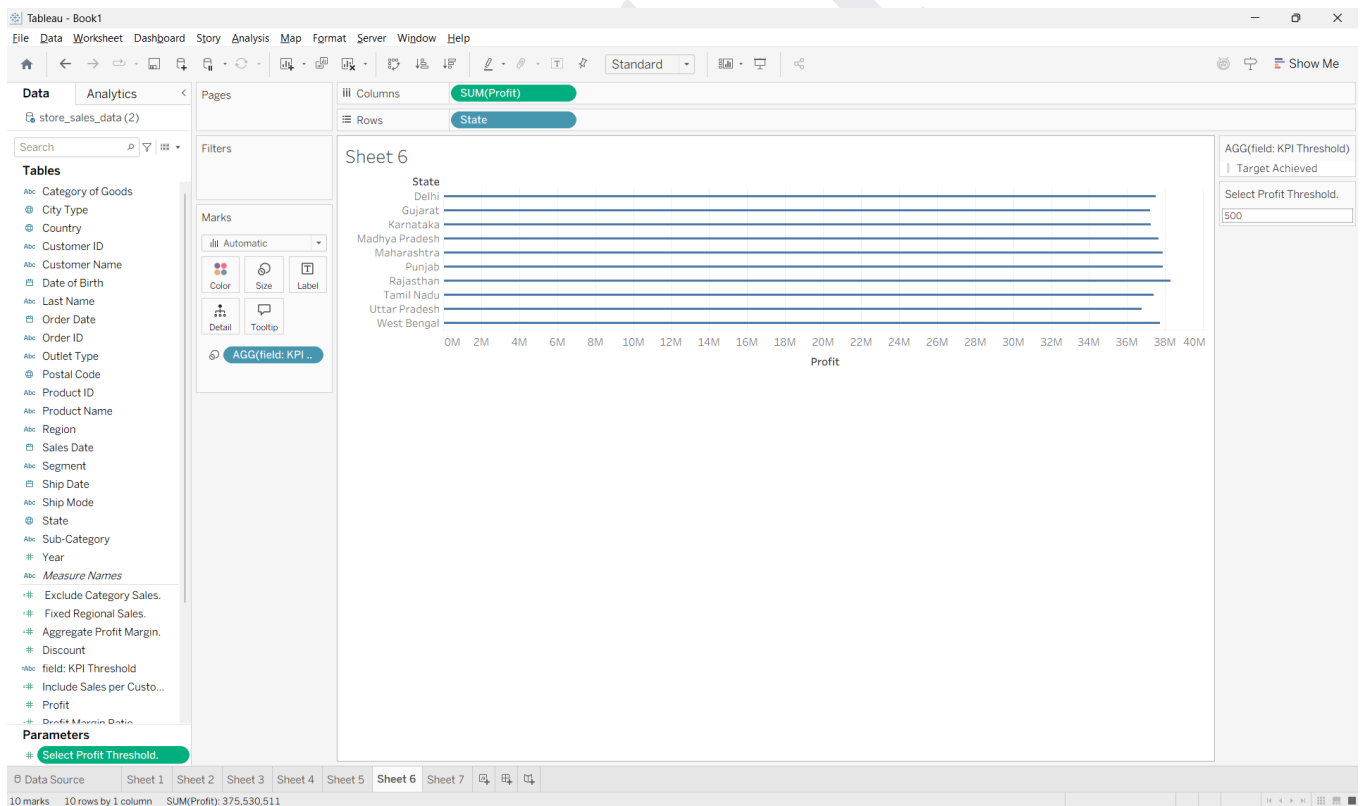
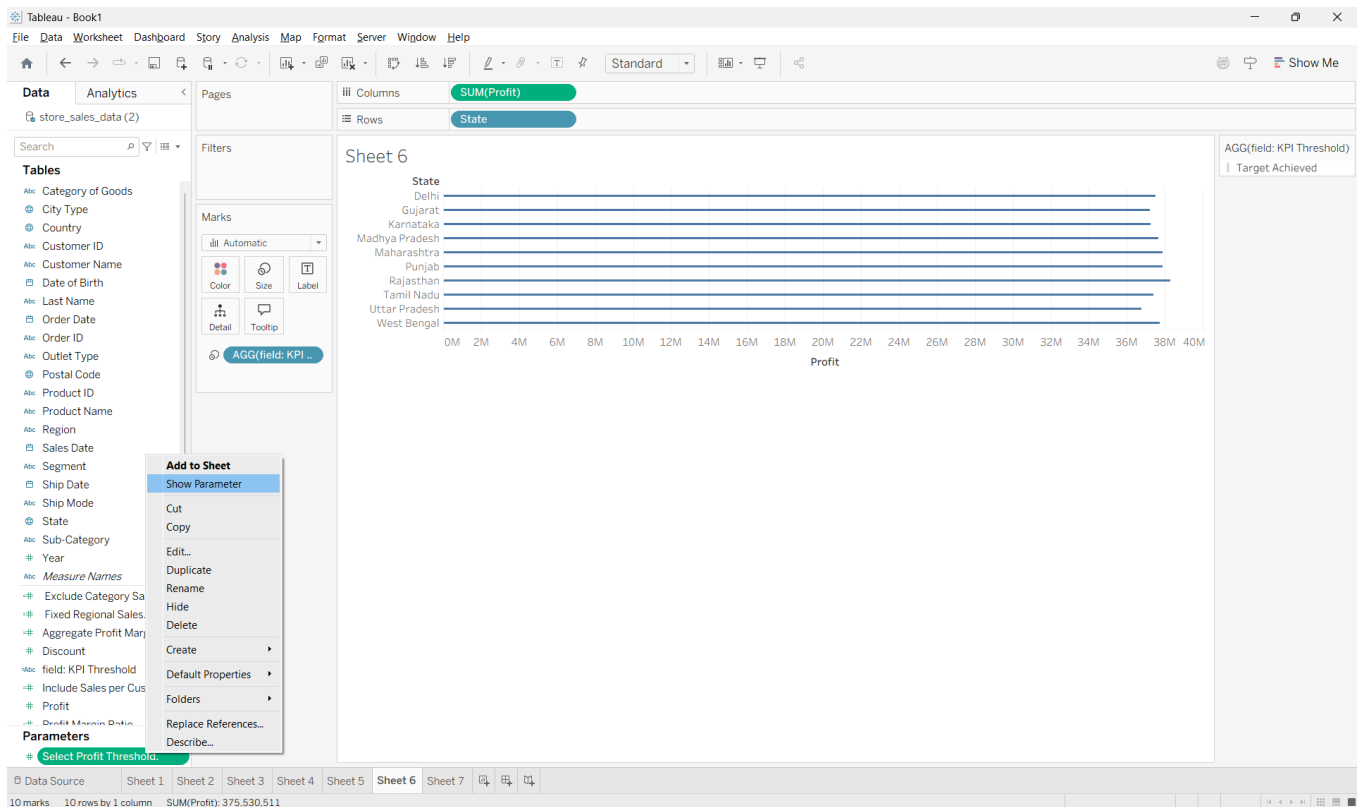
3. Drag the Profit measure and drop it onto the Columns shelf.

4. Drag your newly created calculated field KPI Threshold Met? and drop it directly onto the Color tile in the Marks card.

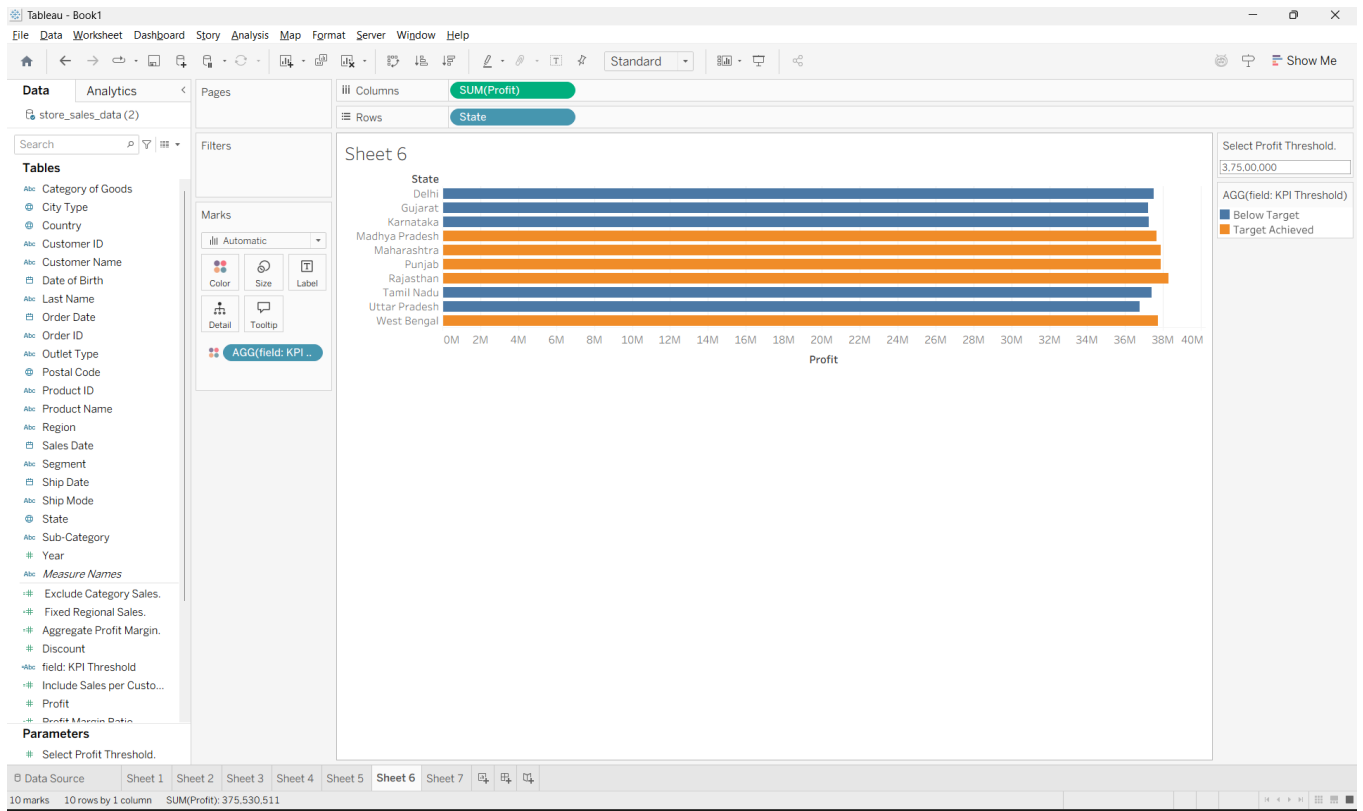


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

4. Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

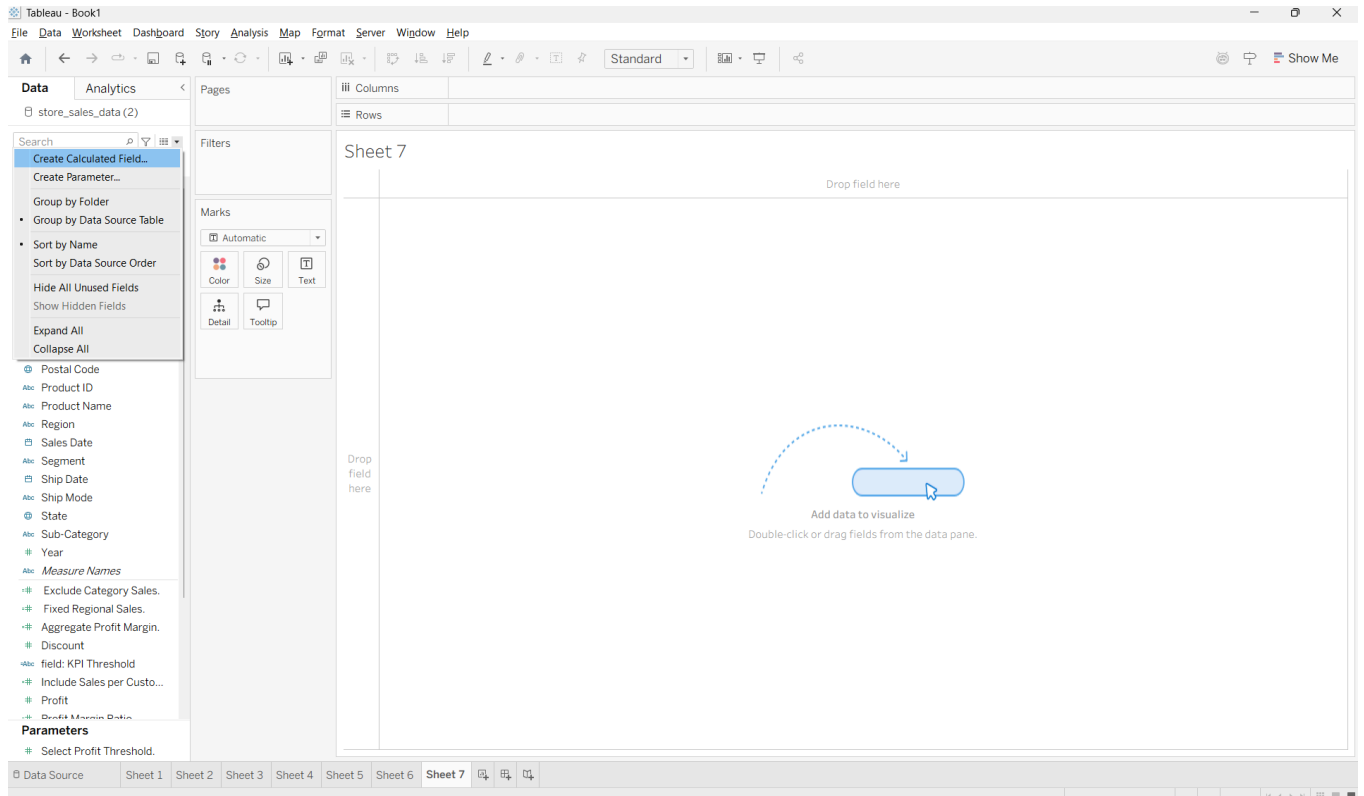
G: Apply Calculations to Fix Data Issues and Enhance Insights

1. Handling Null or Missing Values with IFNULL

In enterprise datasets, missing data or null entries can cause calculations to break or display empty spaces in charts. We can use a logical calculation to substitute a clean value whenever a null occurs.

Let's clean up potential missing indicators in our data:

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

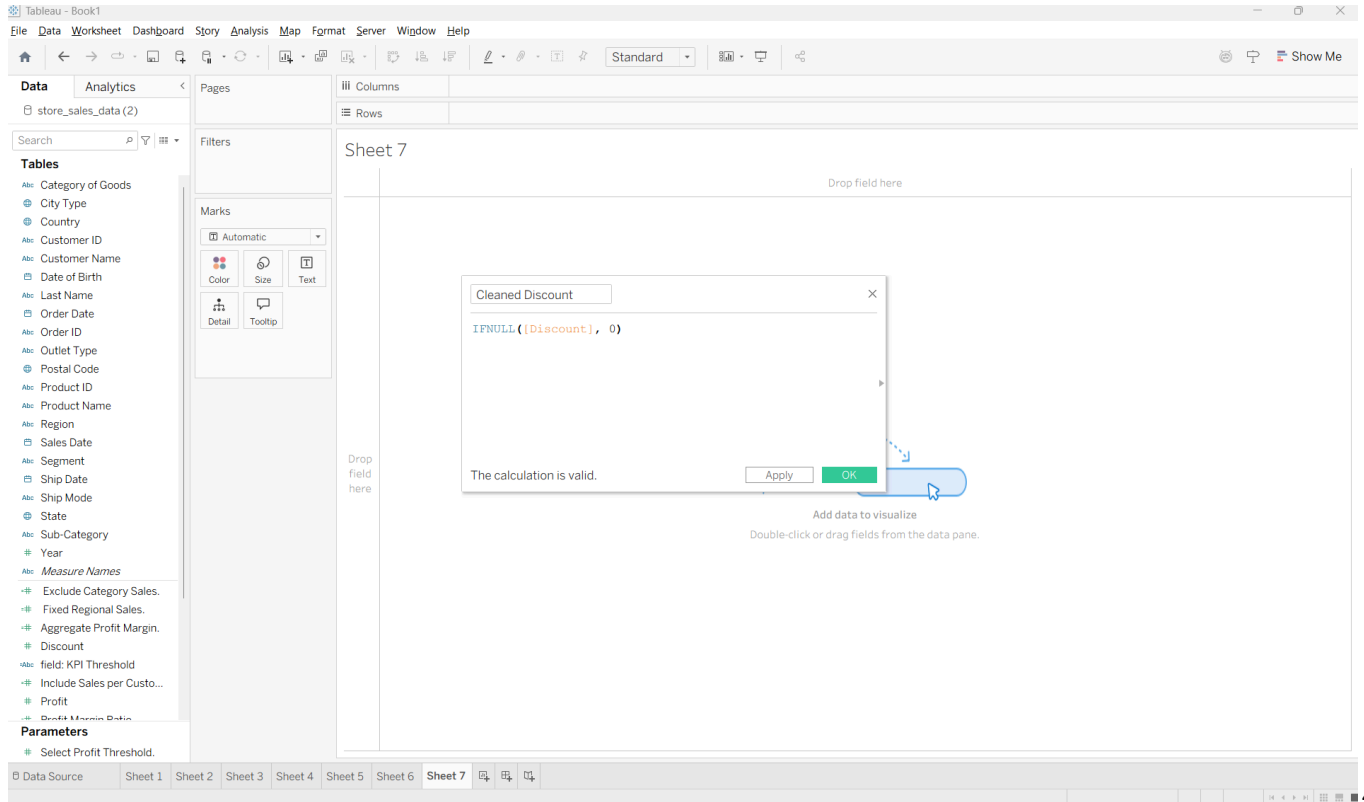


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

2. Name the field: Cleaned Discount.

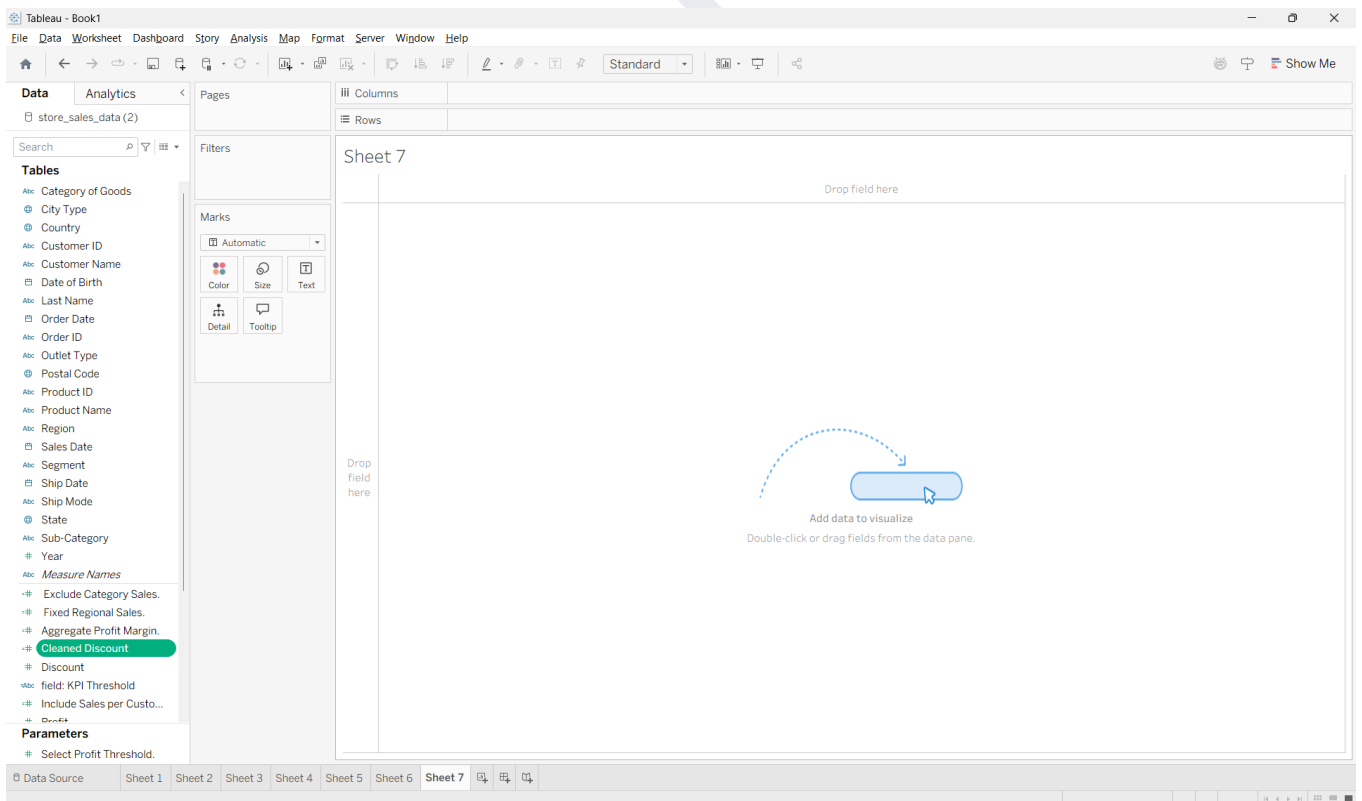
3. Input the following expression:

IFNULL([Discount], 0)



4.

Click OK.



Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

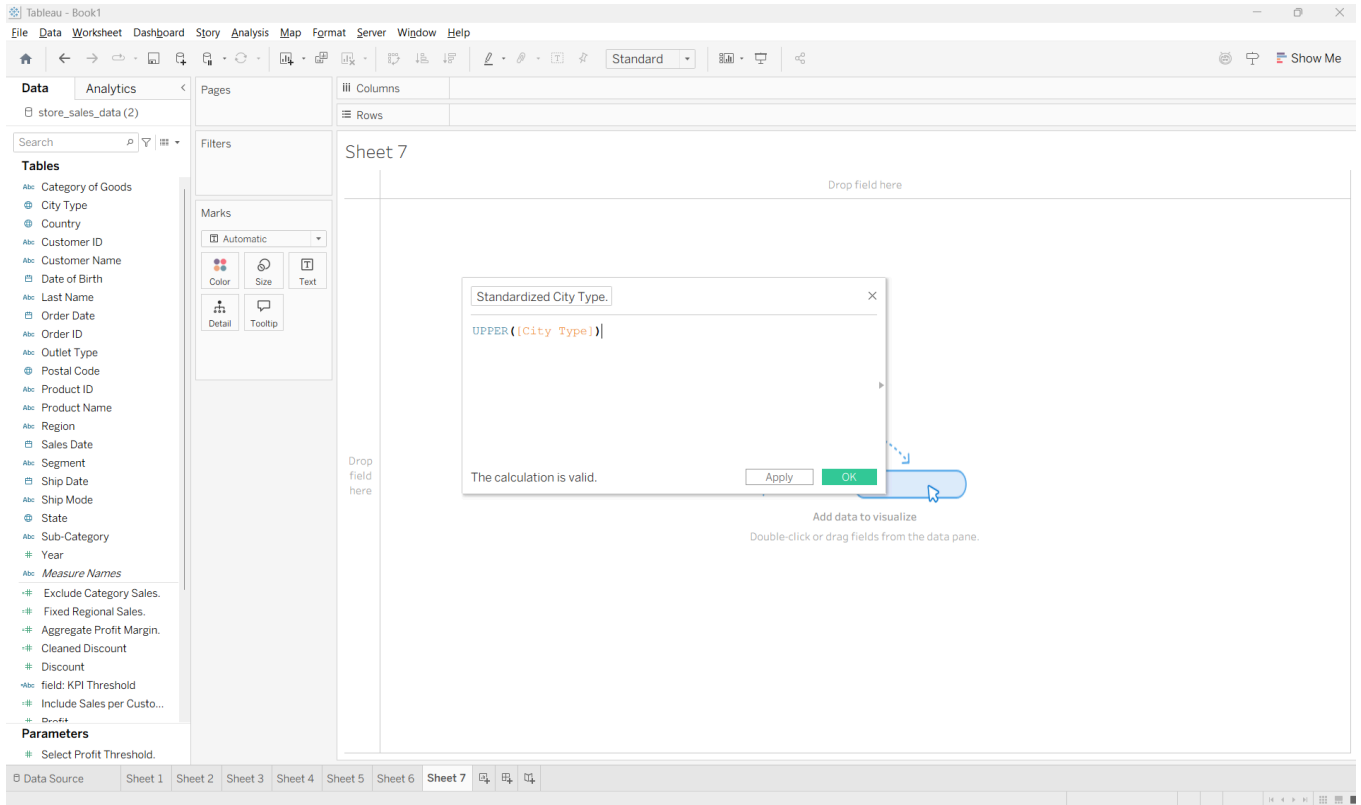
2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

Sometimes database values are entered with inconsistent casing (e.g., "tier 1", "Tier 1", "TIER 1"), which causes Tableau to split them into separate visual blocks. We can force uniform compliance using a string calculation:

1. Create a new Calculated Field named Standardized City Type.

2. Input the following text transformation expression:

UPPER([City Type])

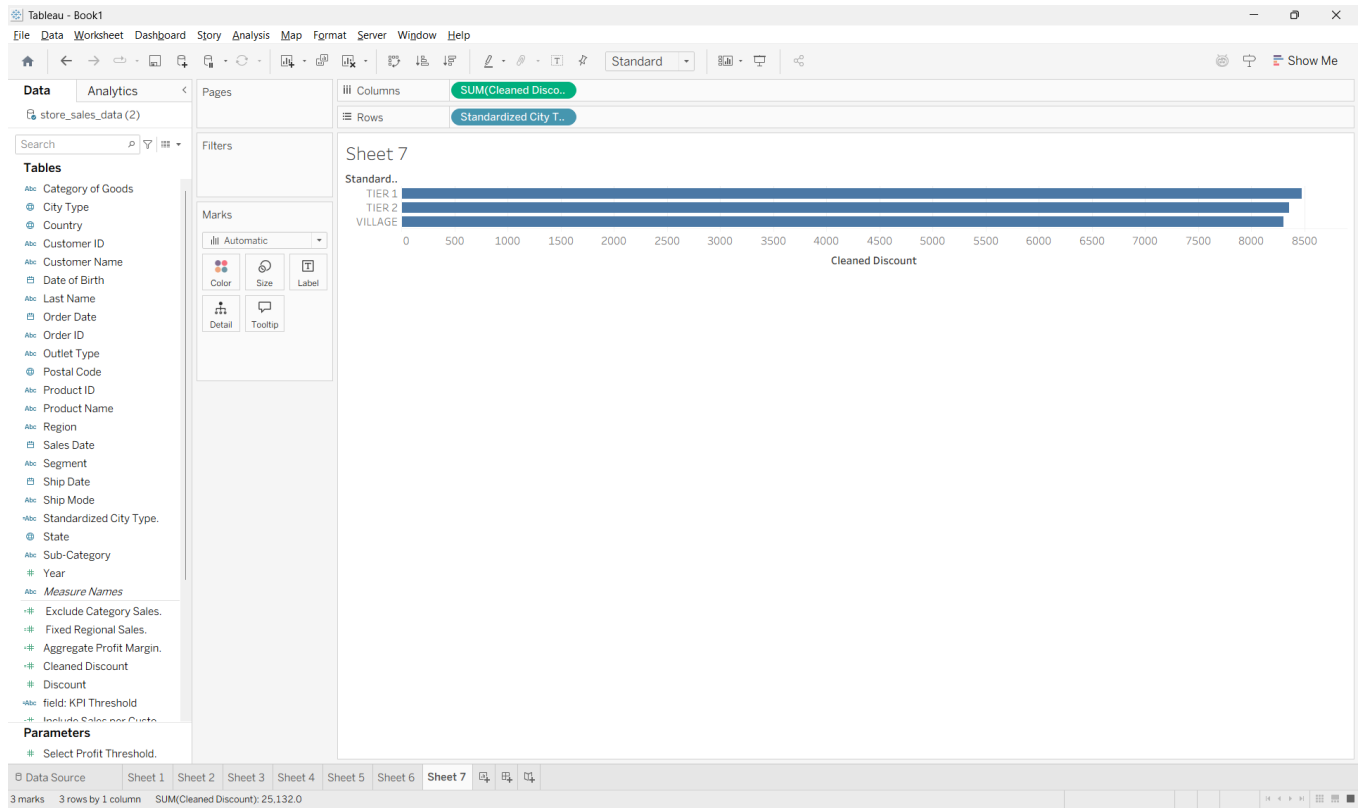


3. Click OK.

Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

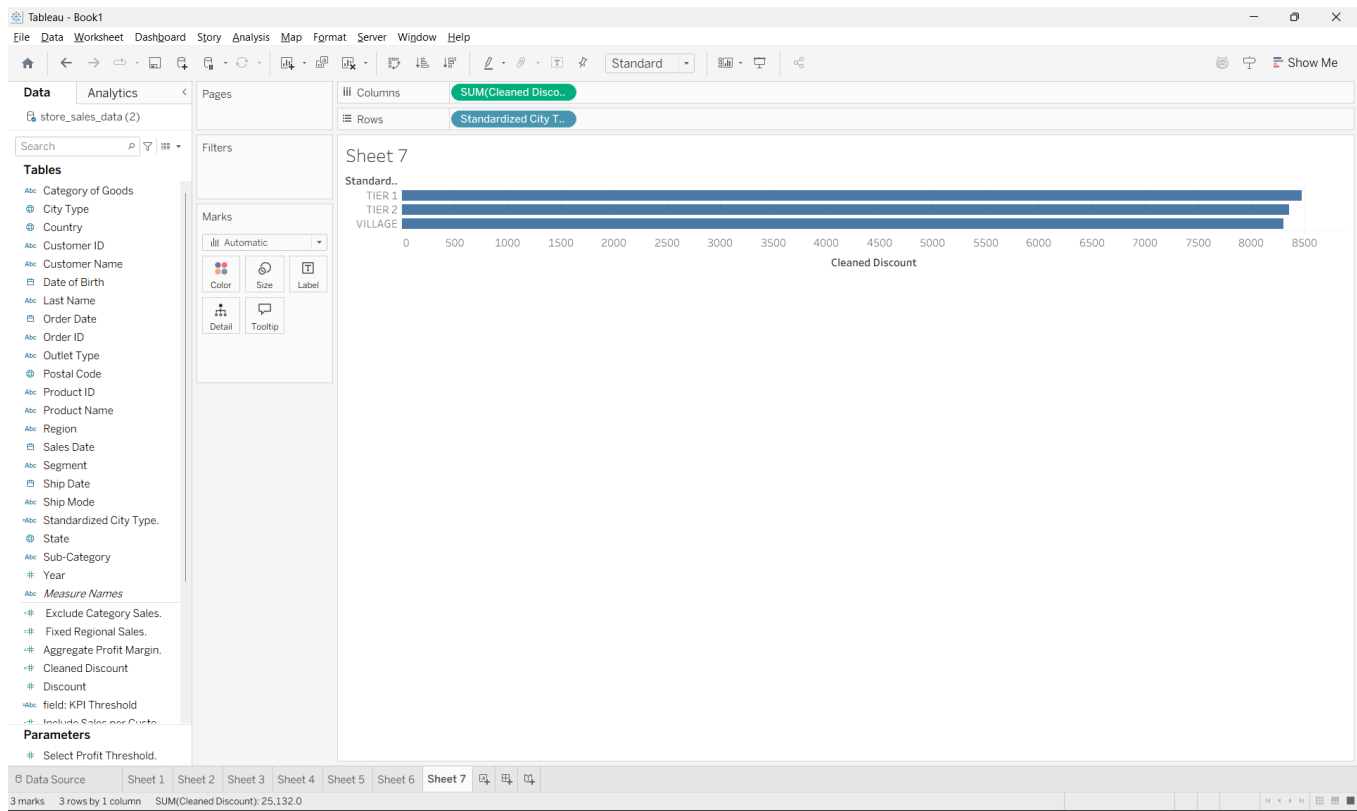
3. Putting the Cleaned Fields to Use

1. Open a new blank worksheet.
2. Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.

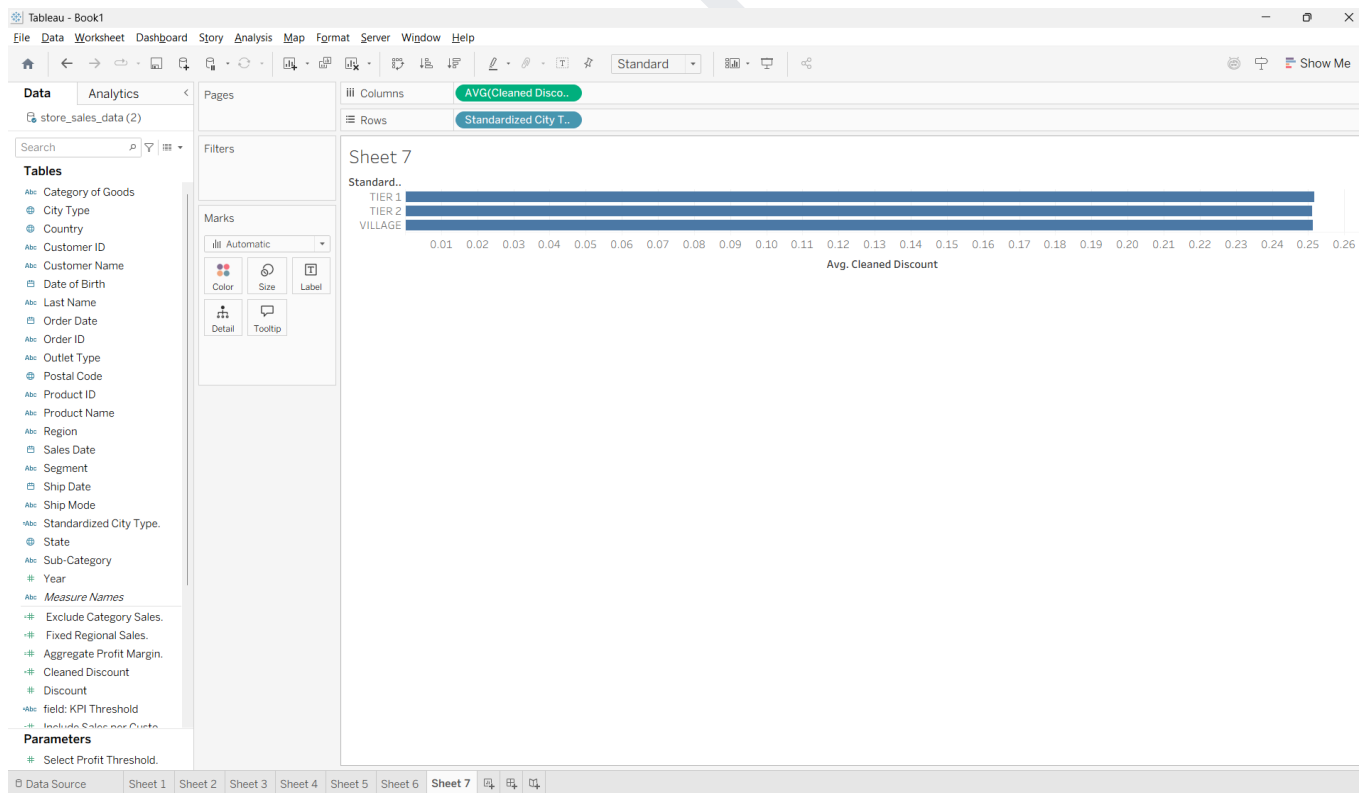


Sheth L.U.J & Sir M.V College
Subject: Data Visualization with Power BI and Tableau

3. Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by



right-clicking the pill.



Sheth L.U.J & Sir M.V College

Subject:Data Visualization with Power BI and Tableau

